

# Lab Images

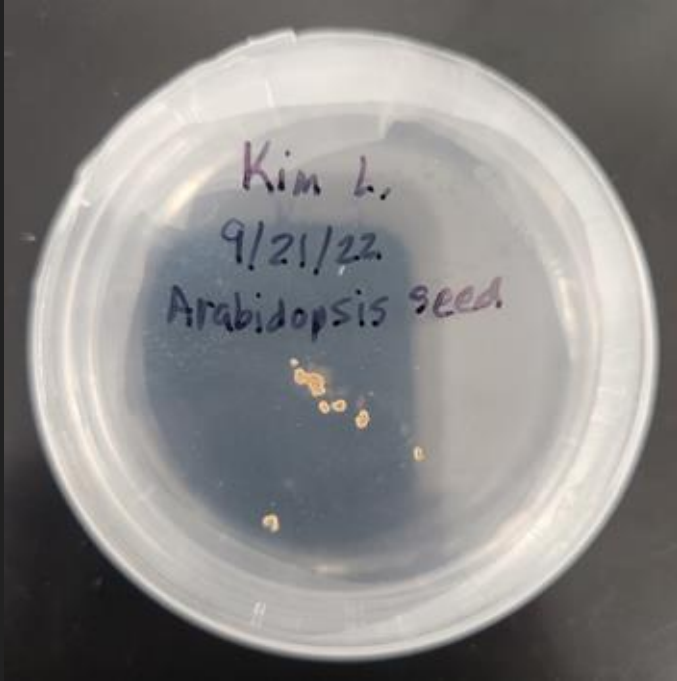
Fall 2022

Kim Luong

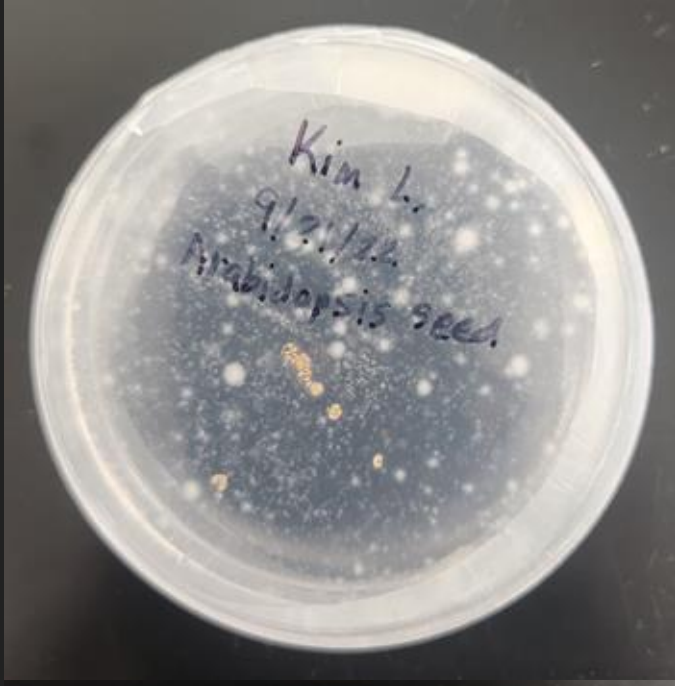
**16 Sept 2022:** Note about using shaker. Place in tighter spot so that it'll hold on. Make sure temperature is set to 20°C.



**21 Sept 2022:** Checking the sterilized Arabidopsis seeds in MS/2 media.



**23 Sept 2022:** After 2 days, the seeds haven't germinated, but there was contamination.



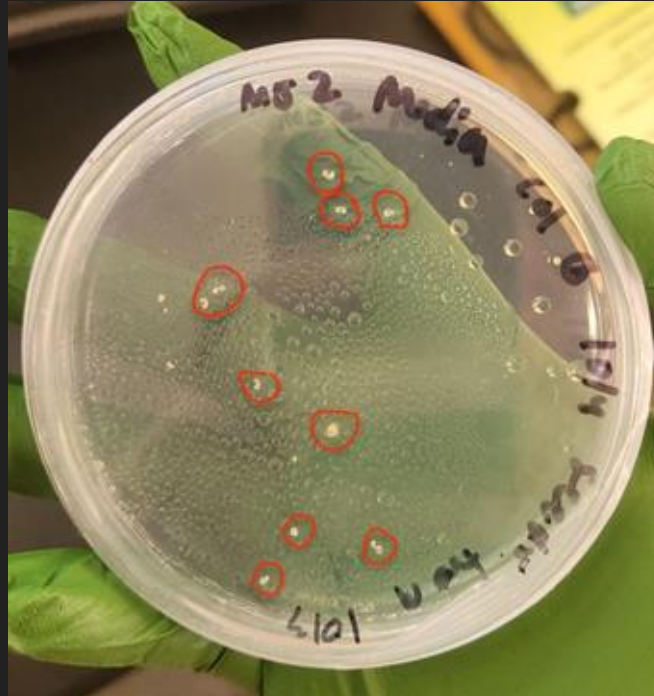
**30 Sept 2022:** Practice taping tip racks. Make sure to press the tape down to make sure that the bottom of the tip racks are sealed and media will no leak.



**6 Oct 2022:** VV sterilized two small batches of seeds on Oct 3rd. Germination plates have been plated with MS/2 media on Oct 4th.

- Observation: After 2 days, some of the seeds have started to germinate (circled).  
11/15 of the planted seeds germinated → 73.33% germination rate

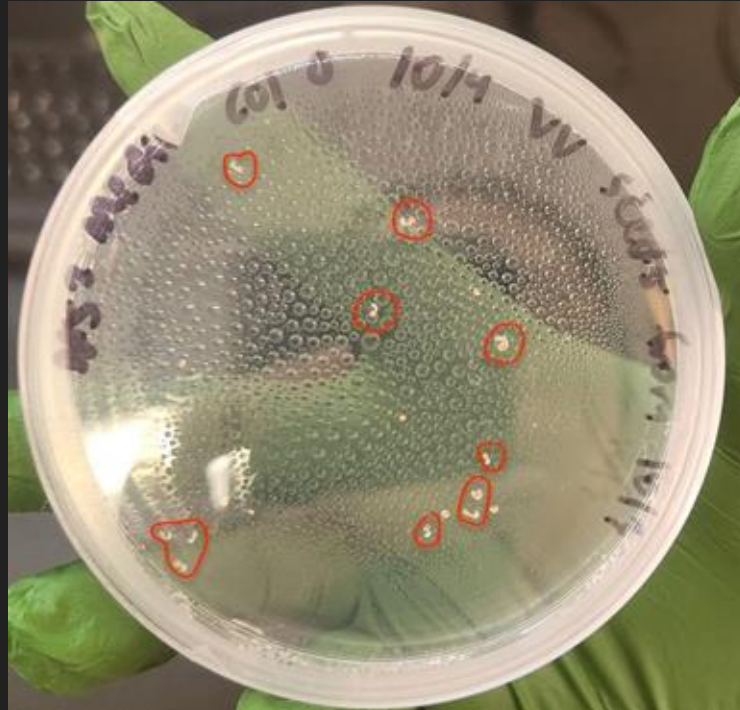
Note: Seeds will be located in the pink seed container labeled sterile in green tape in the growth room.



**6 Oct 2022:** VV sterilized two small batches of seeds on Oct 3rd. Germination plates have been plated with MS/2 media on Oct 4th.

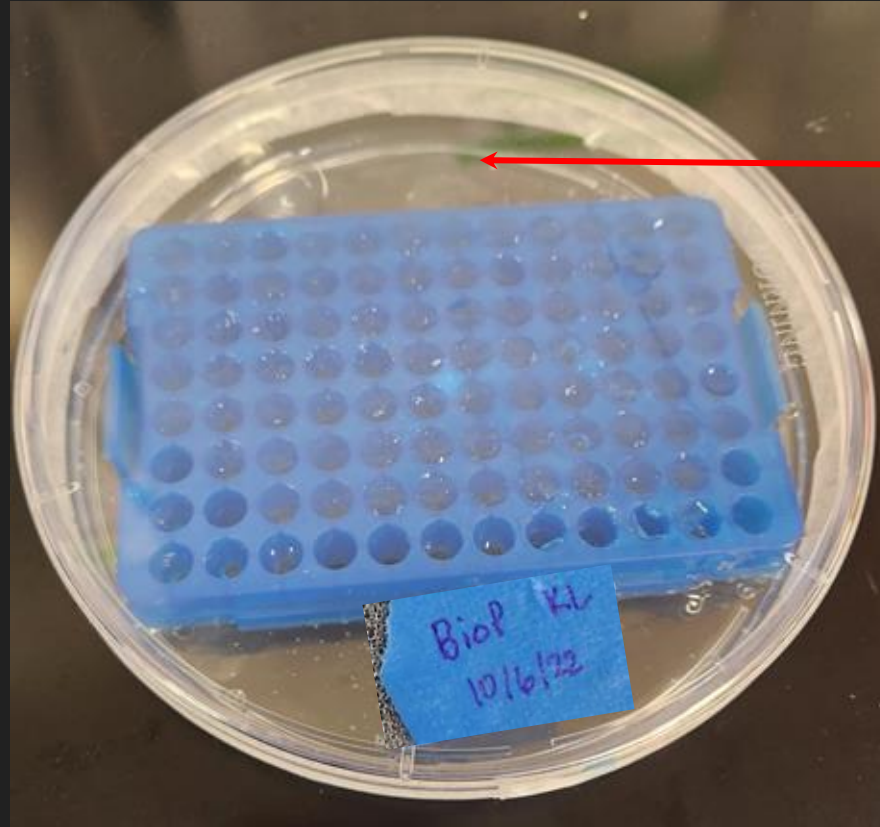
- Observation: After 2 days, some of the seeds have started to germinate (circled)  
11/18 of the planted seeds germinated → 61.11% germination rate.

Note: Seeds will be located in the pink seed container labeled sterile in green tape in the growth room.



## 6 Oct 2022: First hydroponic set up (control group).

- Compared to hydroponics from other students, John's medium solidified after a couple of minutes when poured into the petri plates instead of remaining as a liquid.

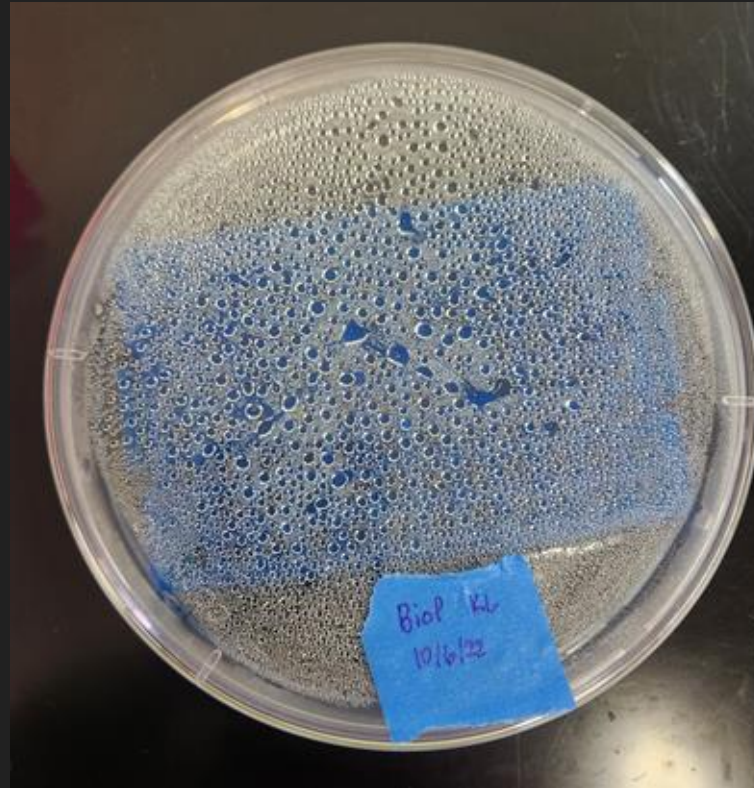


Solidified John's medium when it is supposed to stay as a liquid.

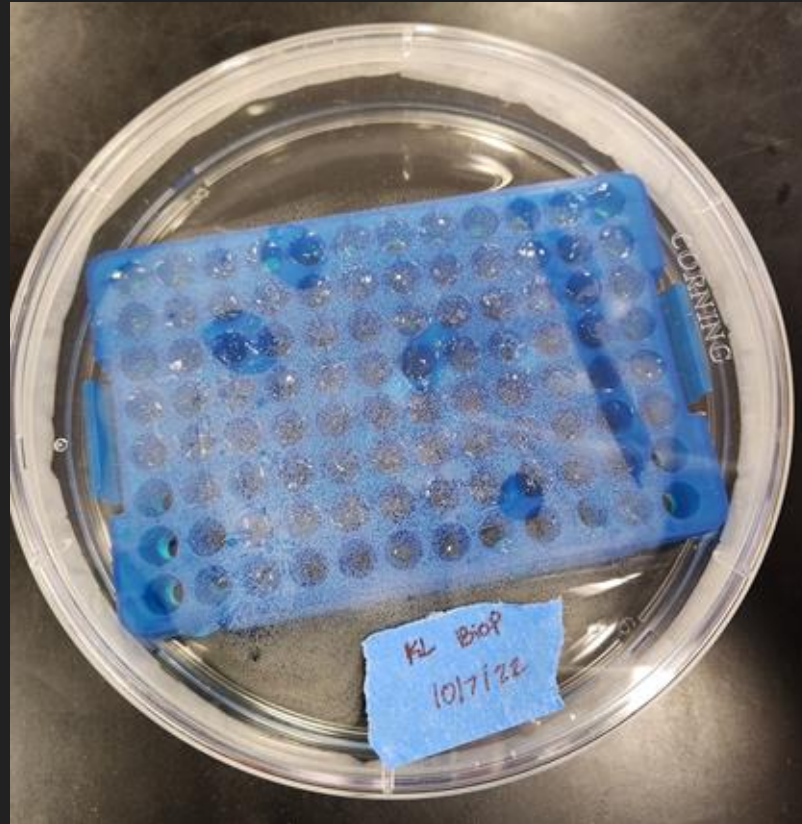
Note: The autoclaved tip racks, surgical tape, and petri plates are located in the drawer called "Plant Stuff".

Biop KL  
10/6/22

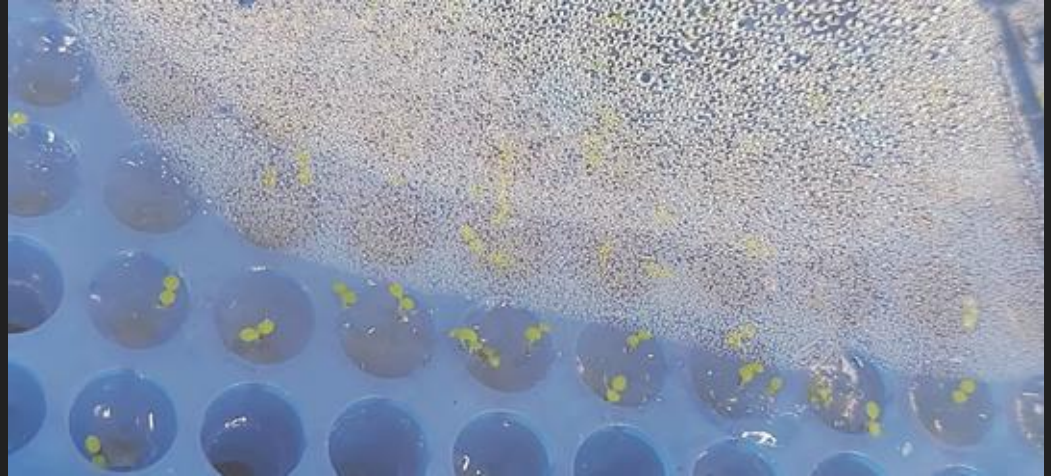
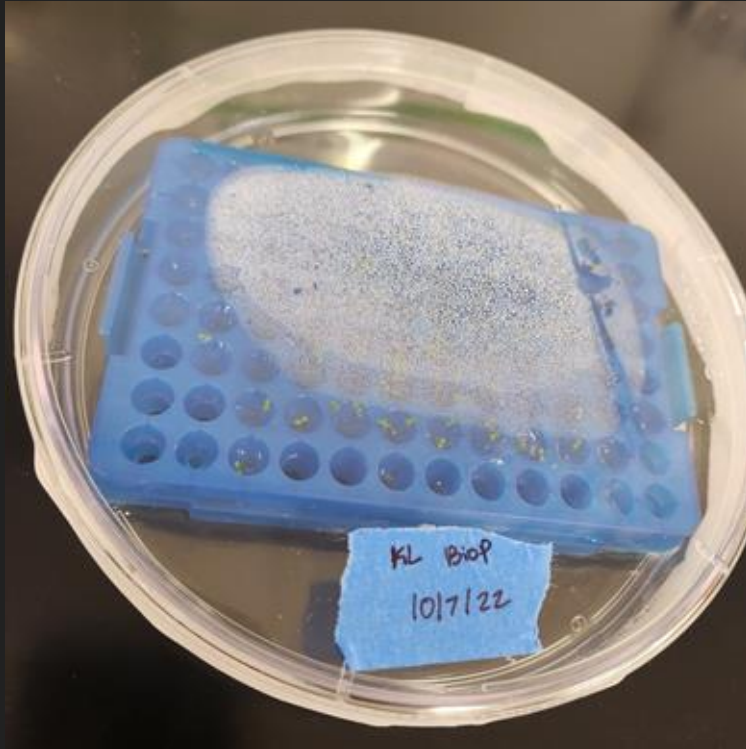
**7 Oct 2022:** First hydroponic set up (control group). After 1 day, the seeds have not germinated. I realized that the reason that medium solidified in the petri plate was that I used John's medium that already had agar prepared in it.



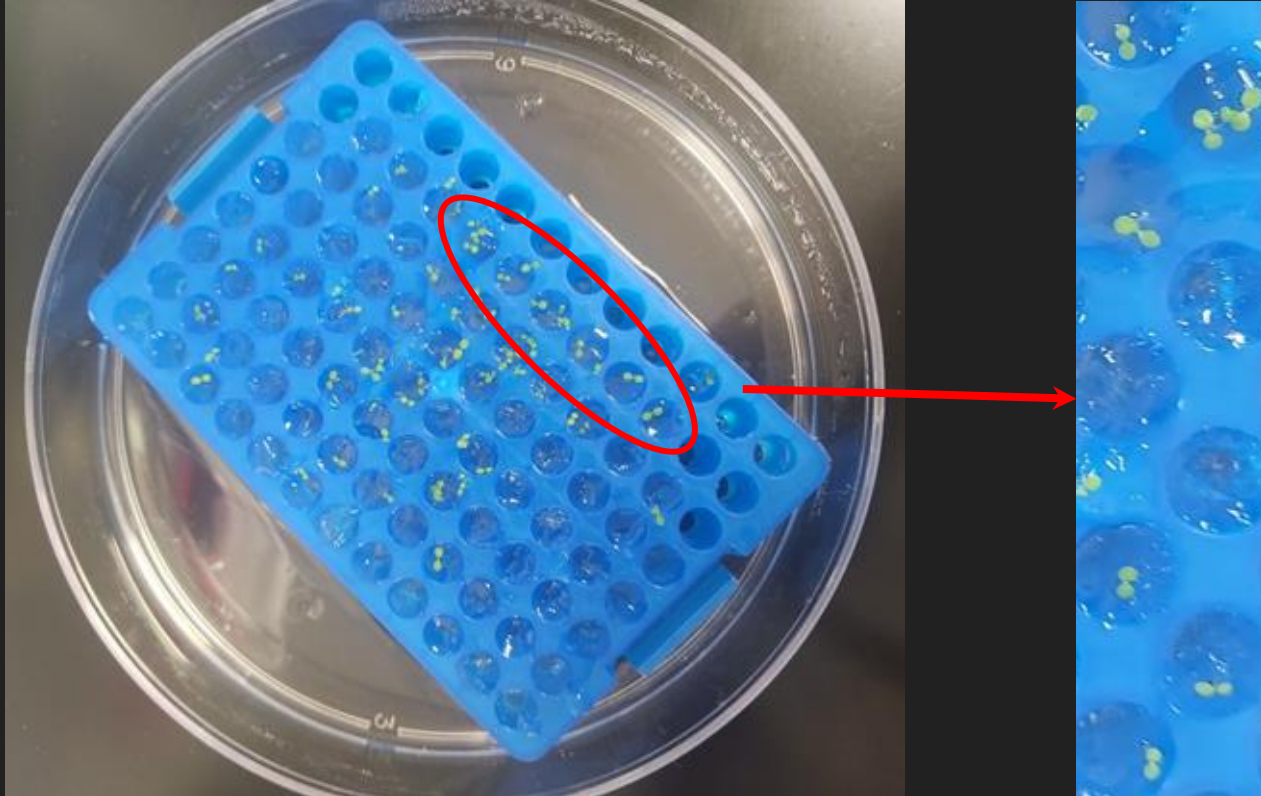
**7 Oct 2022:** Second hydroponic set up (Control A). This time the medium that was in the tip rack solidified, which enabled me to place seeds in. A different bottle of John's medium was used to fill the plate with 20 ml of John's medium. This time the medium remained as a liquid.



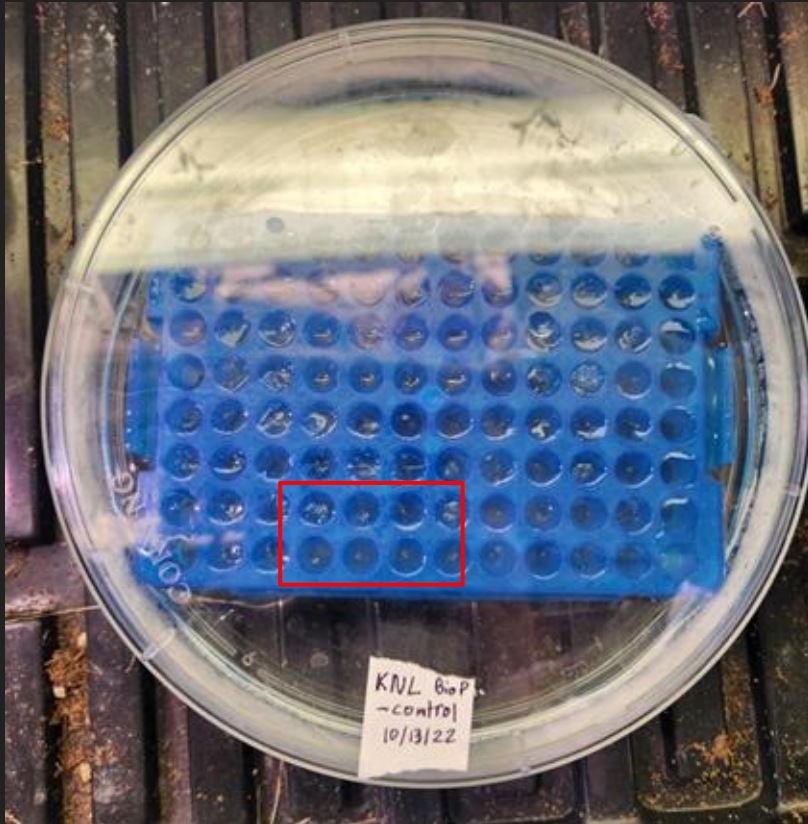
**11 Oct 2022:** After four days, the *Arabidopsis thaliana* seeds have germinated in the hydroponic system with the negative control (Control A). However, almost all of the medium in the petri plate has evaporated. As a result, 20 ml of John's medium is not enough for the *Arabidopsis* seedlings for 2 weeks.



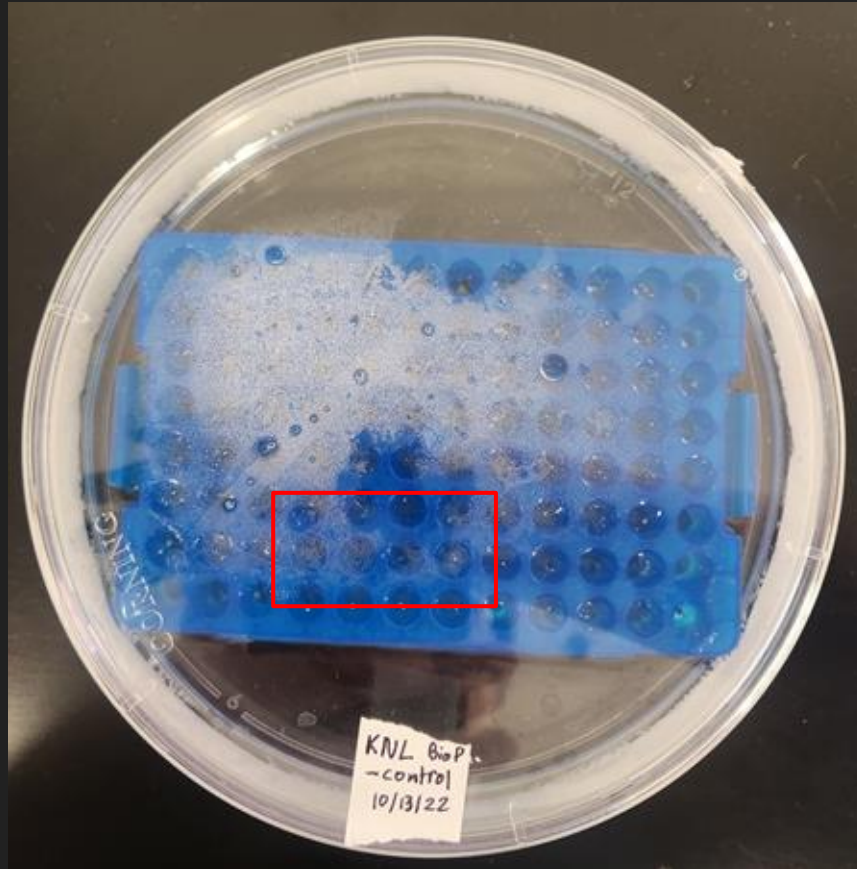
**12 Oct 2022:** Control A: After 5 days, all of the medium has evaporated. Thus, the seedlings look dehydrated, in which the cotyledon do not look firm, the seedlings aren't completely upright, and the seedlings also started to have a light/pale green color.



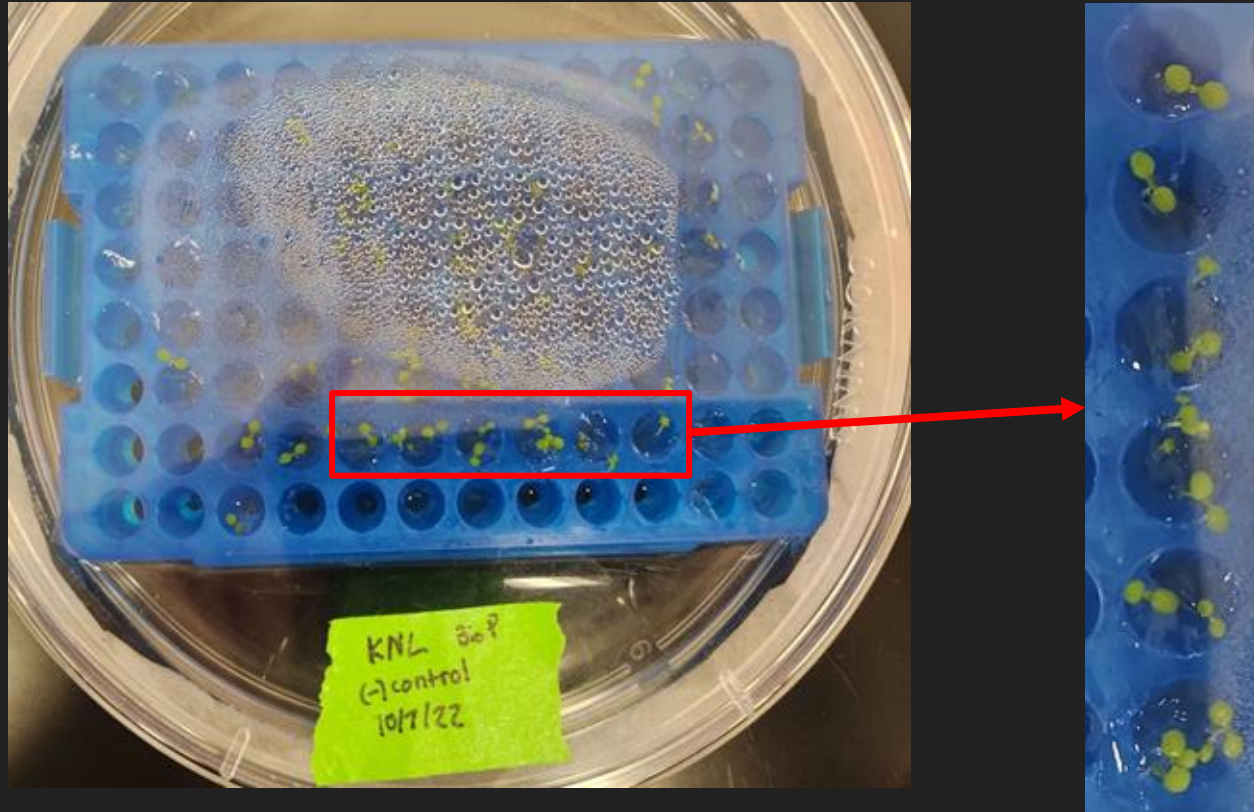
**13 Oct 2022:** Third hydroponic set up. (Control B - Replica for negative control) The hydroponic this time looks a lot cleaner/better than the previous one. This was done with the updated protocol. This time seeds were planted evenly as possible throughout majority of the agar plugs.



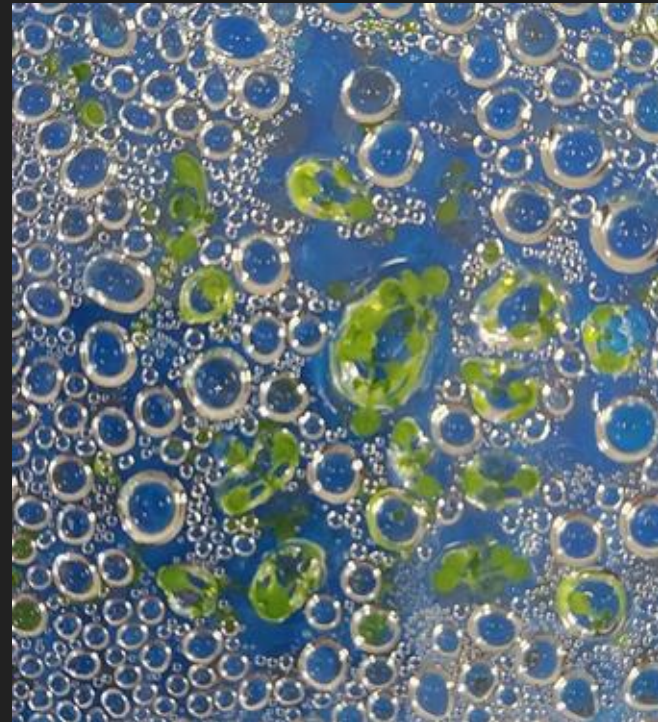
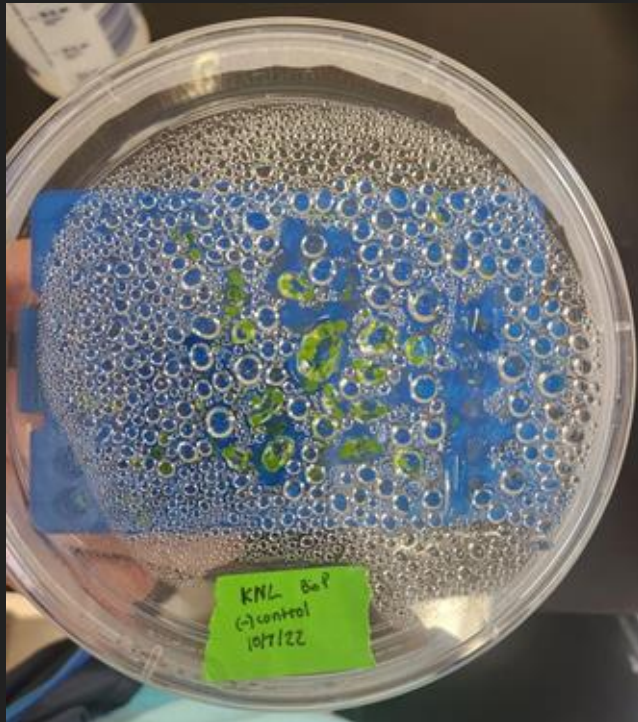
**14 Oct 2022:** After 1 day of Control B, there is no germination.



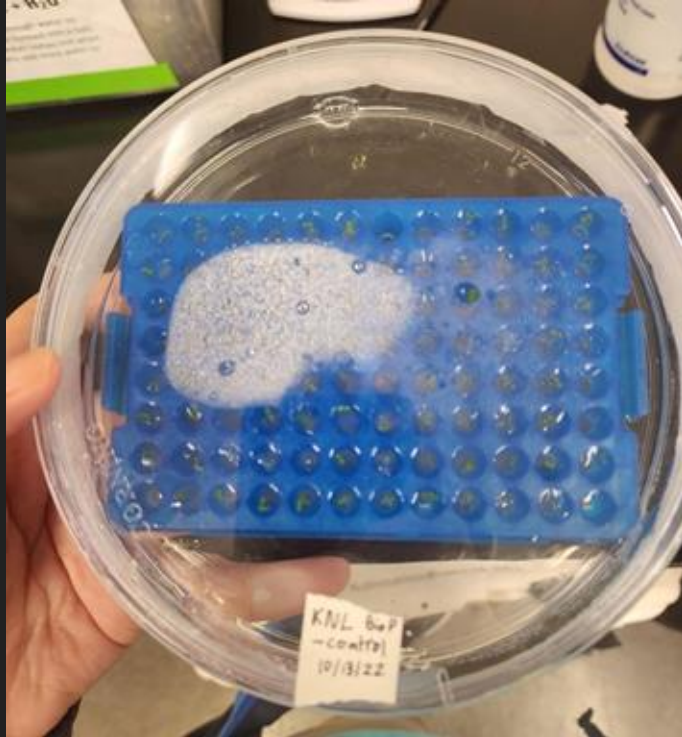
**14 Oct 2022:** After 2 days of the medium being added to Control A, the seedlings look a little bit more healthier, in which the seedlings have a darker green color.



**17 Oct 2022:** Control A - The seedlings have grown a lot over the weekend. Some of the leaves are touching the roof of the petri plates. In fact, these seedlings are wilting and look very wet. This probably indicates that the seedling is receiving too much moisture, since they are touching the roof of the plates where the media has concentrated. Although it hasn't been two weeks yet, the seedlings should be transferred to food containers.



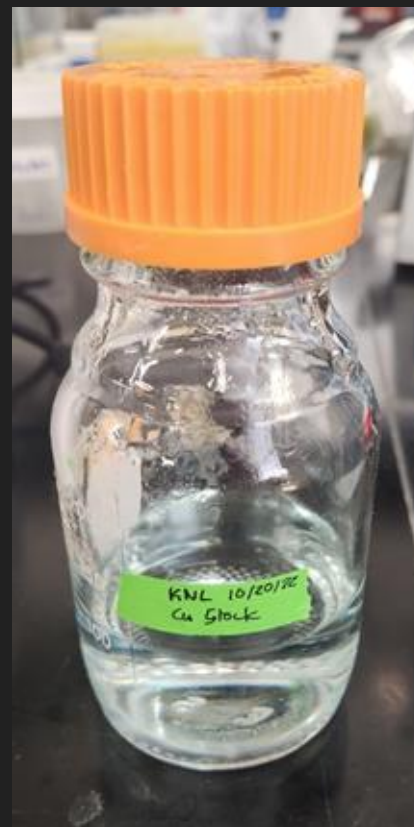
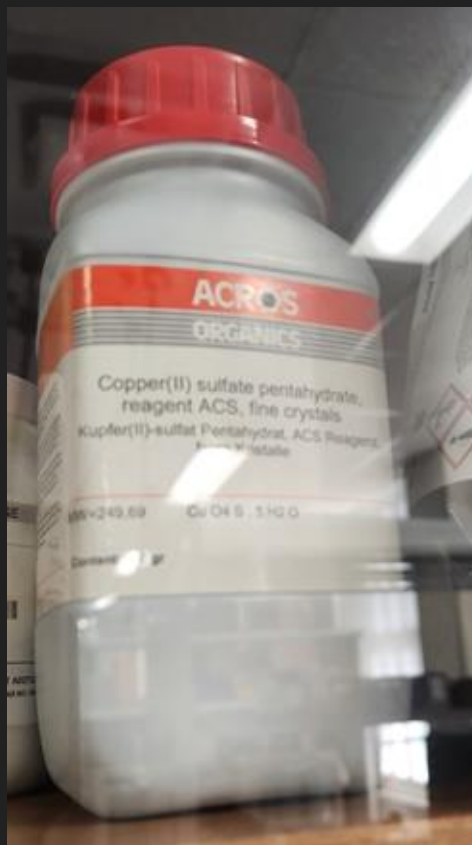
**17 Oct 2022:** After 4 days, the Arabidopsis seeds have germinated in Control B.



**20 Oct 2022:** After 2 weeks, Control A was moved to the food container. Instead of pouring John's media up to the middle of the tip rack, I should pour the media up to the bottom of the tip rack.

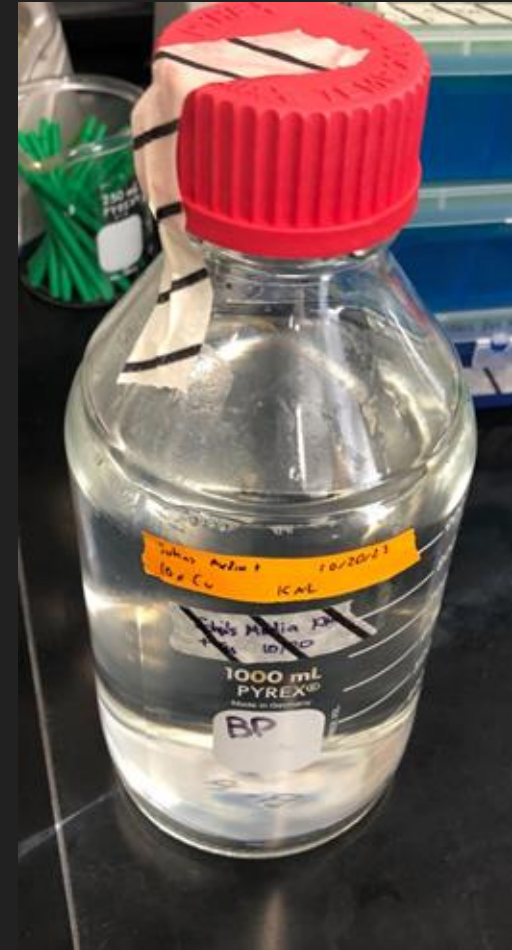


**20 Oct 2022:** Made Copper Stock Solution with 186.74 mg of Copper (II) Sulfate Pentahydrate and 100 ml of DI Water. (Make sure that the copper reagent dissolves completely.)

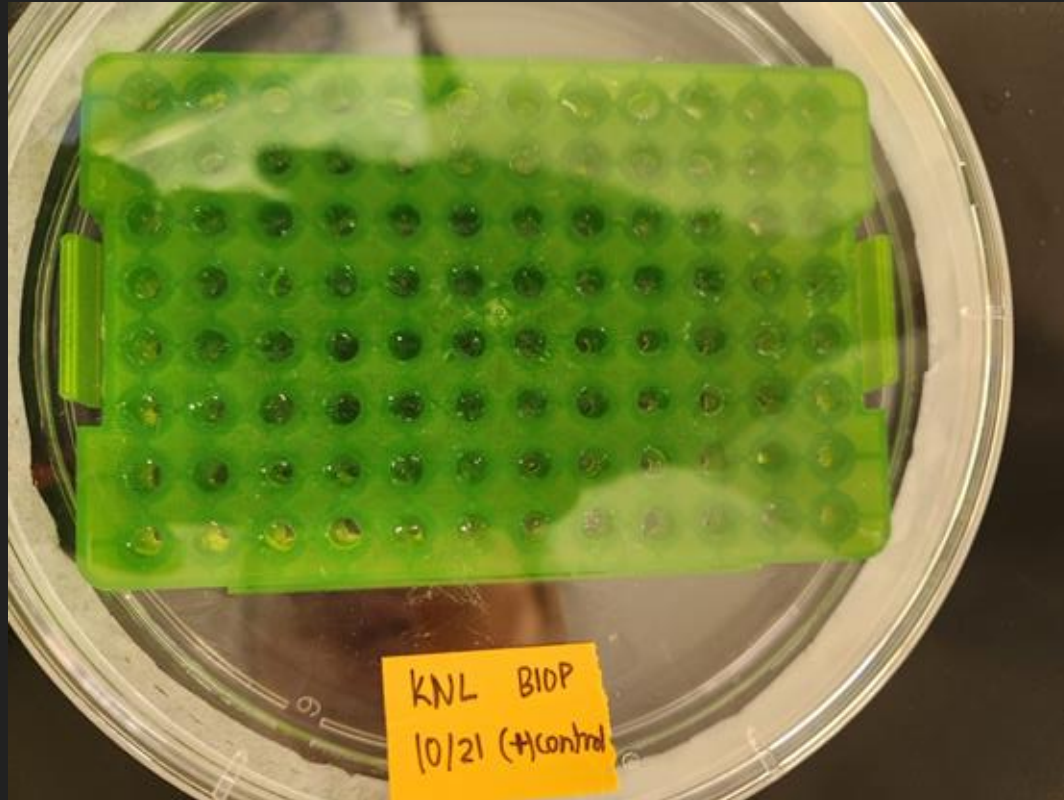


**20 Oct 2022:** Made John's media for positive control group (excess copper).

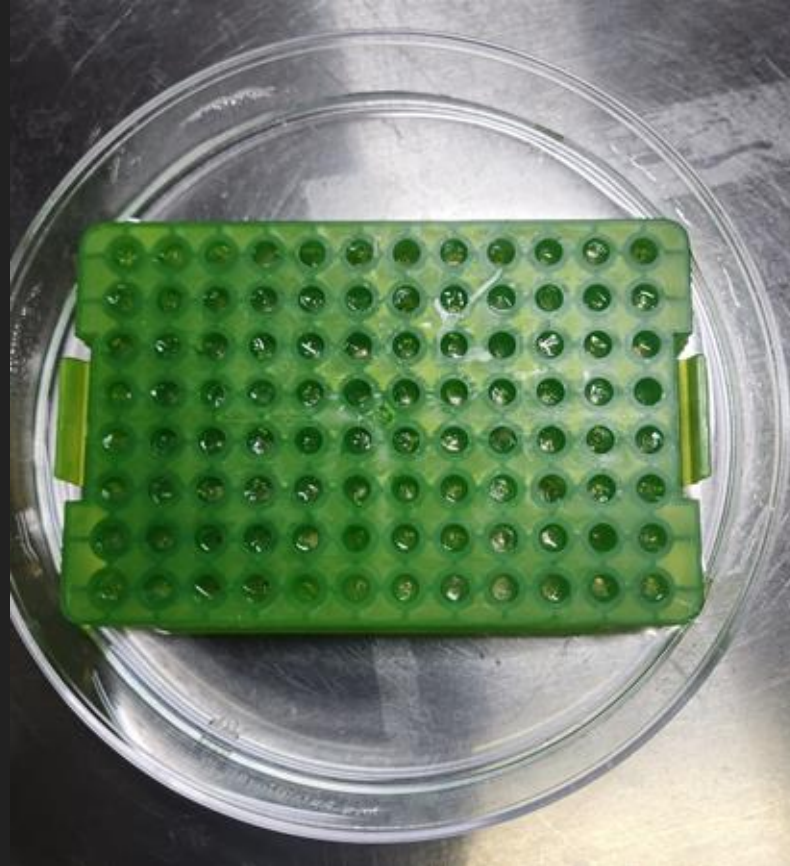
- $\text{CaSO}_4$  is the only thing that needs to be measured on the scale.
- Separate micropipette tips should be utilized for 1 ml of  $\text{NH}_4\text{NO}_3$ , 1 ml of NaFe EDTA, 1 ml of 1000x microelements, 1 ml of Copper Stock solution, and 4 ml of  $\text{KH}_2\text{PO}_4 + \text{MgSO}_4 + \text{H}_2\text{O}$ .
- Fill up the rest of the bottle to 1 L with DI Water.



**21 Oct 2022:** New hydroponic set up was created for positive control (Experimental A). The seedlings in this hydroponic will be exposed to excess copper for 1-2 days.



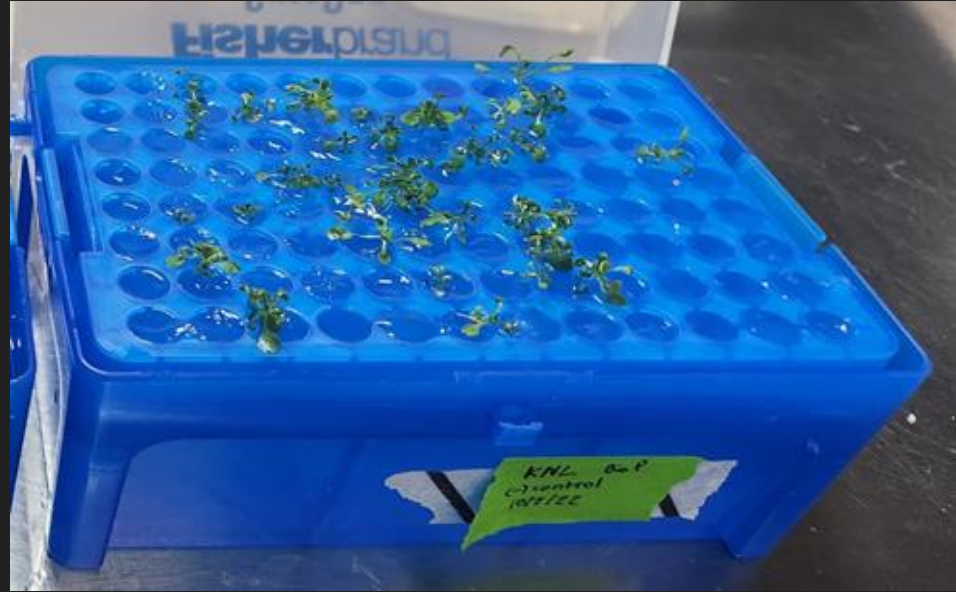
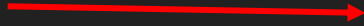
**24 Oct 2022:** After 3 days, the seedlings in Experimental A are starting to germinate and grow.



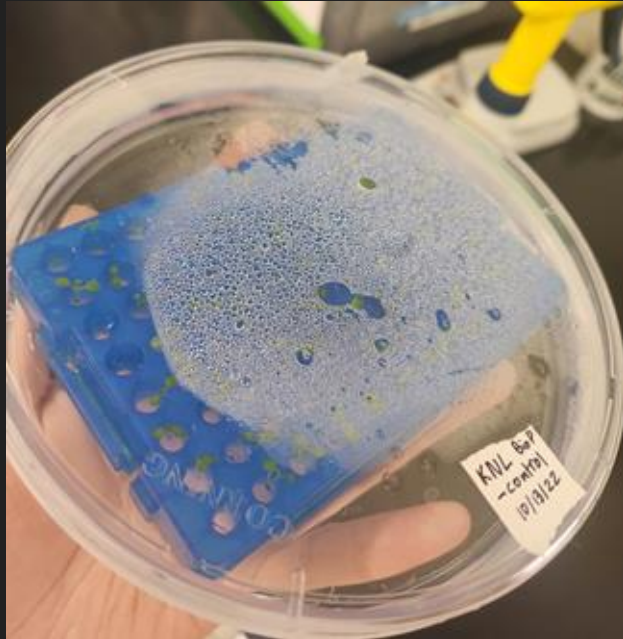
**24 Oct 2022:** Control B - the seedlings are touching the top of the petri plates. As a result, to avoid the seedlings having too much moisture from the condensation of the John's media, the lid was lifted slightly. This hydroponic system will be moved to a tip box tomorrow.



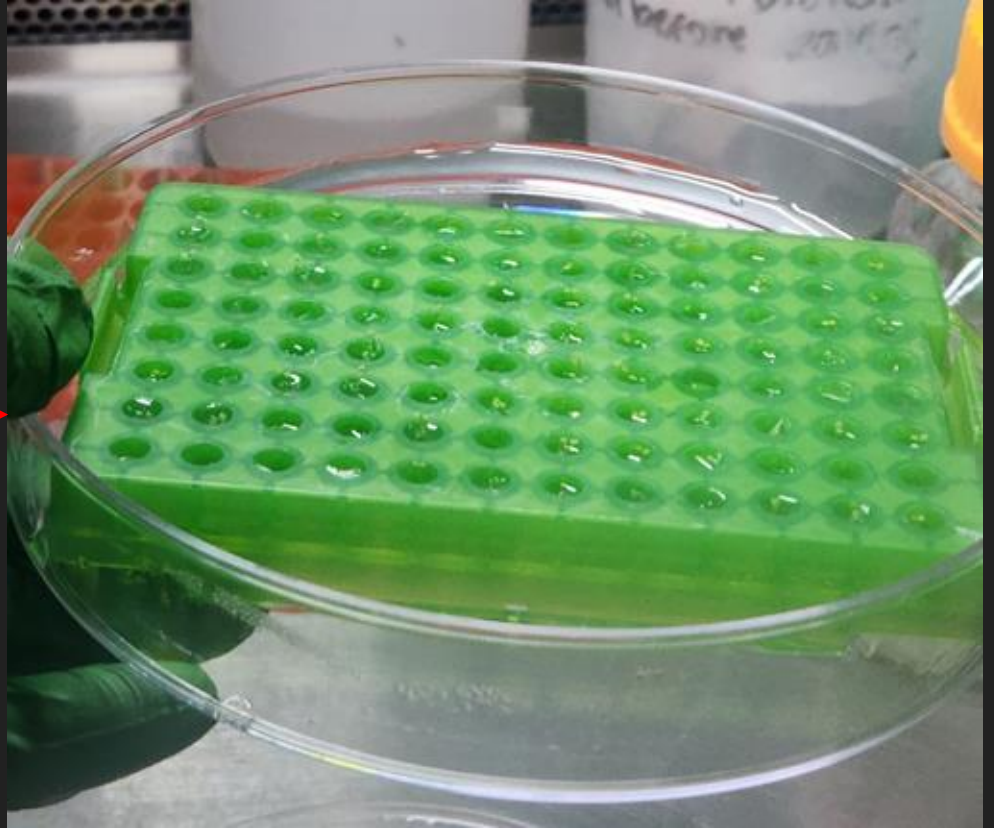
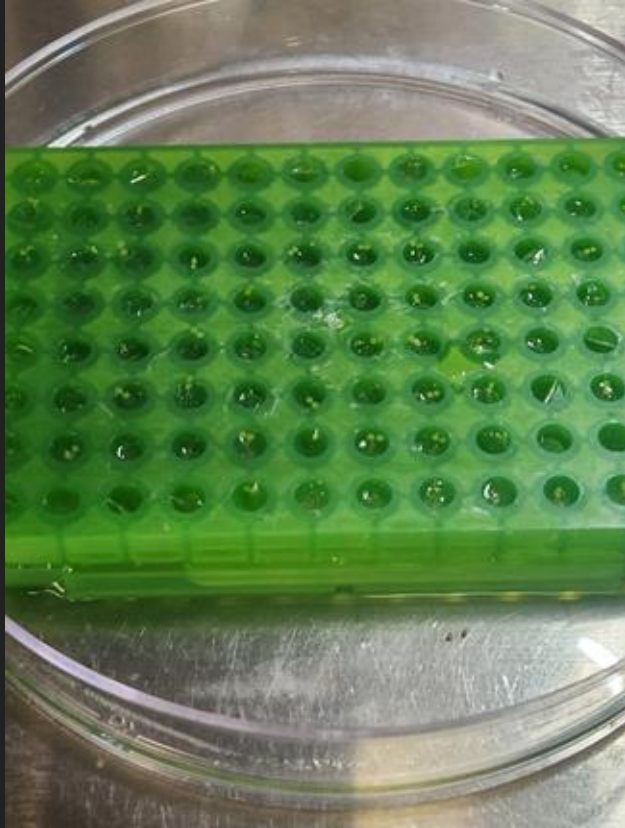
**25 Oct 2022:** Due to the changes in the protocol, Control A is transferred from the food container to the tip boxes.



**25 Oct 2022:** After approximately 2 weeks, the tip rack from Control B is transferred to the tip box instead of the food container.



**25 Oct 2022:** Experimental A was exposed to John's media with excess copper for 3 days (left). The media in the plate was replaced with normal John's media (right).



**25 Oct 2022:** New hydroponic set up was created for positive control (Experimental B). The seedlings in this hydroponic will be exposed to excess copper for 3 days.



## 28 Oct 2022: Control A (Day 21)

- Some of the plants, especially the ones that were affected from being cramped and over watered under the petri plates, are drying up and have a pale green/yellowish color. In addition, some of the wells that consist of the agar John's media is drying up.



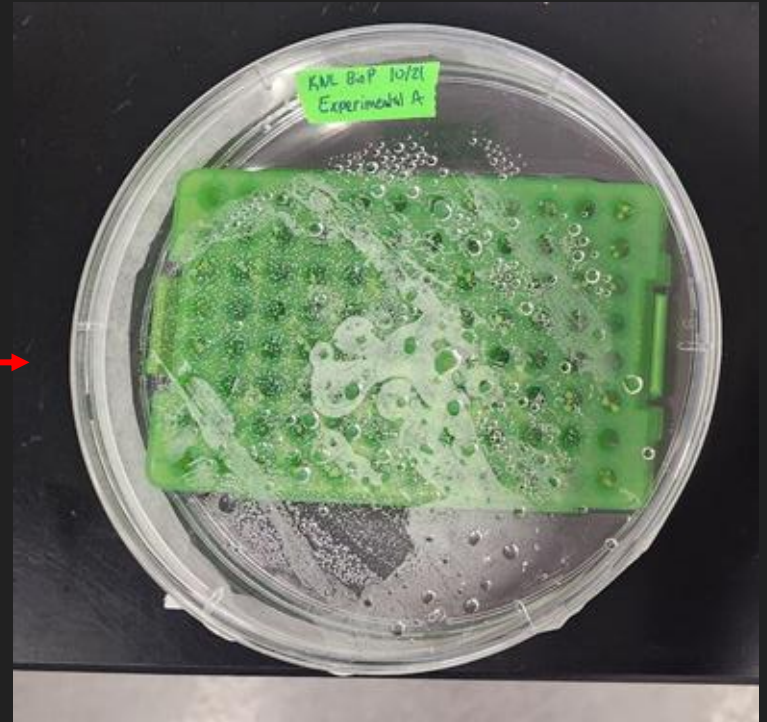
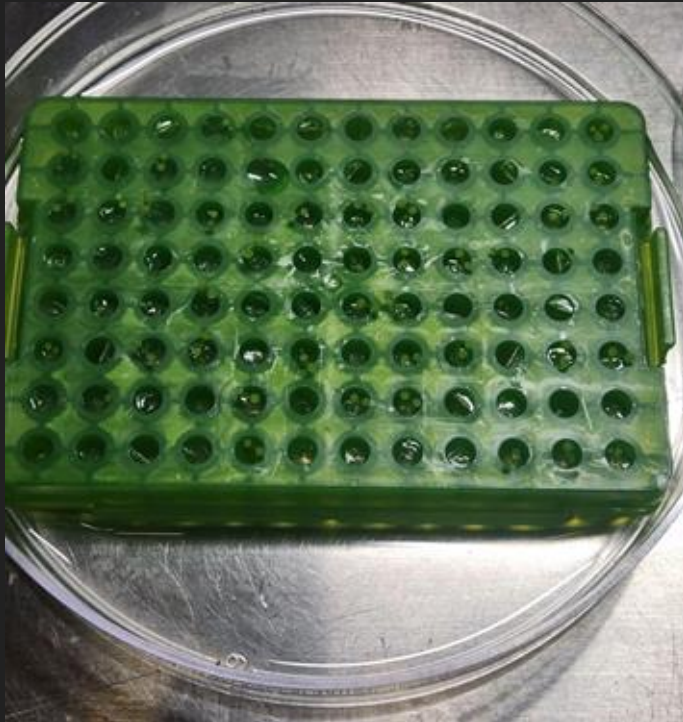
## 28 Oct 2022: Control B (Day 15)

- The plants in Control B look far more healthier, in which the plant is taller and the color of the plant is green. There are a couple of plants that are wilting, especially the ones that just started growing. Although most of the agar remains moist, some of the agar are dried up.



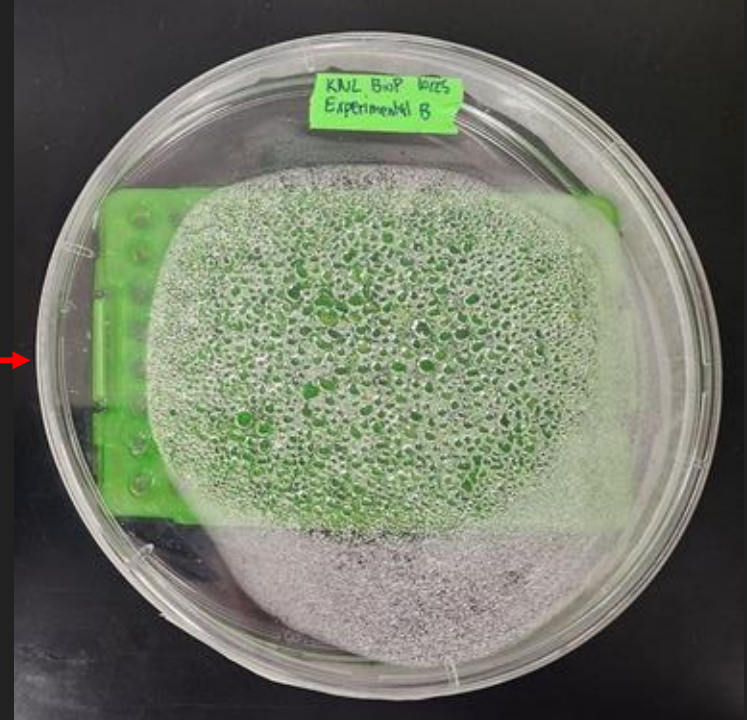
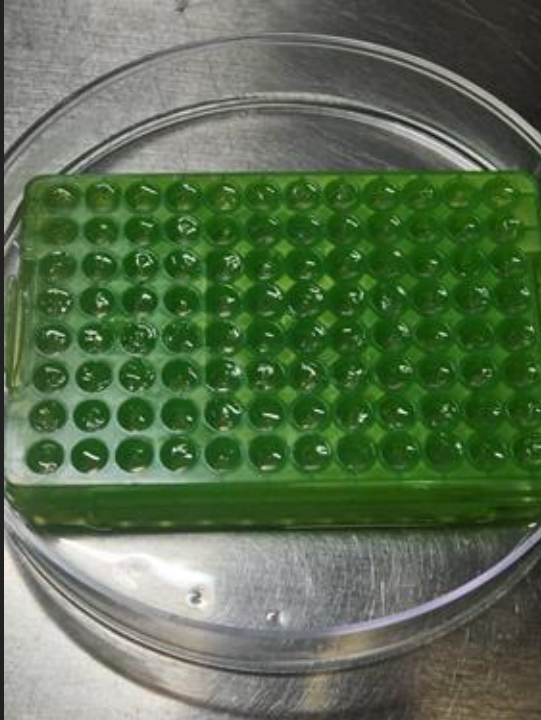
## 28 Oct 2022: Experimental A (Day 7)

- The liquid media is changing from normal John's media (left) to John's media with excess copper (right).



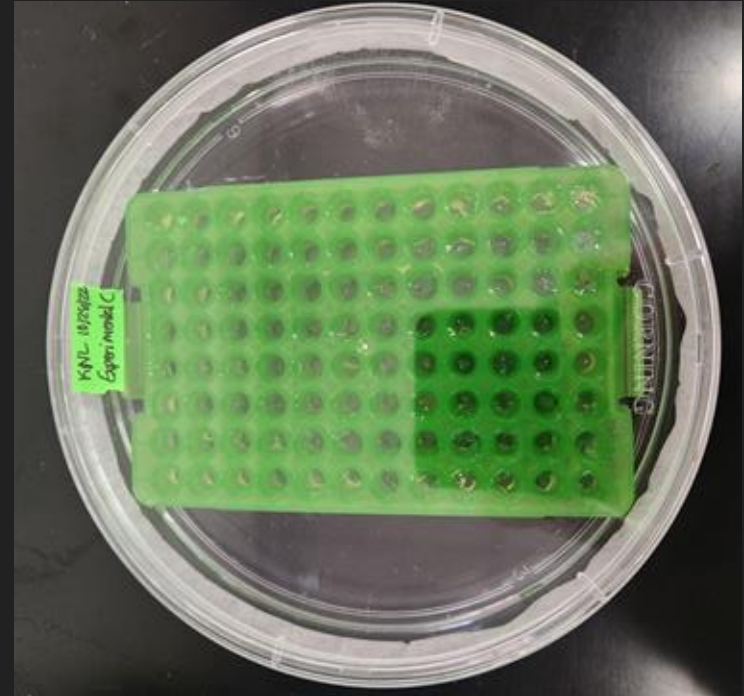
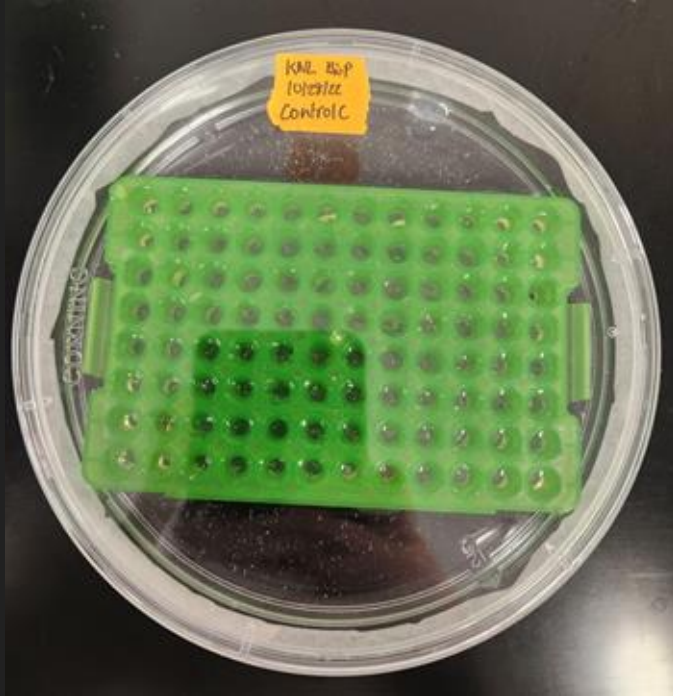
## 28 Oct 2022: Experimental B (Day 3)

- The liquid media is changing from John's media with excess copper (left) to normal John's media (right).



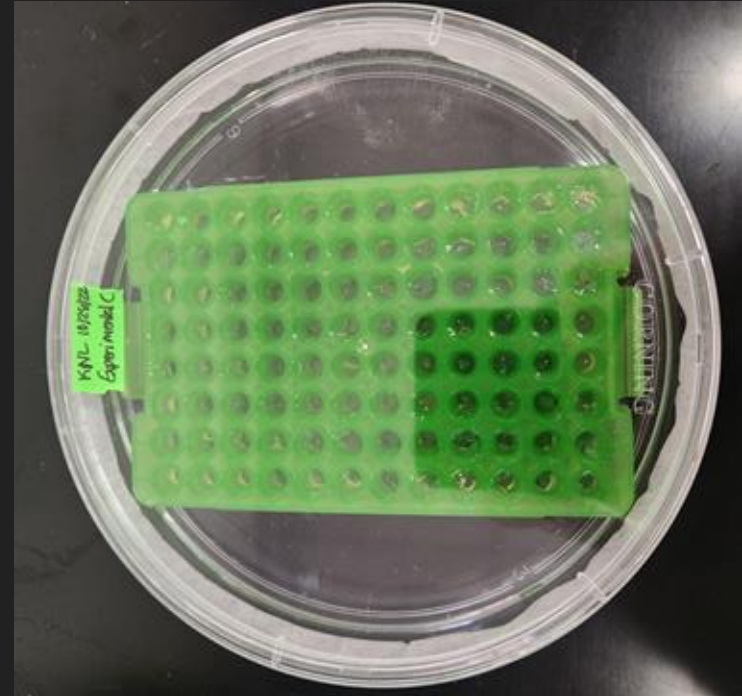
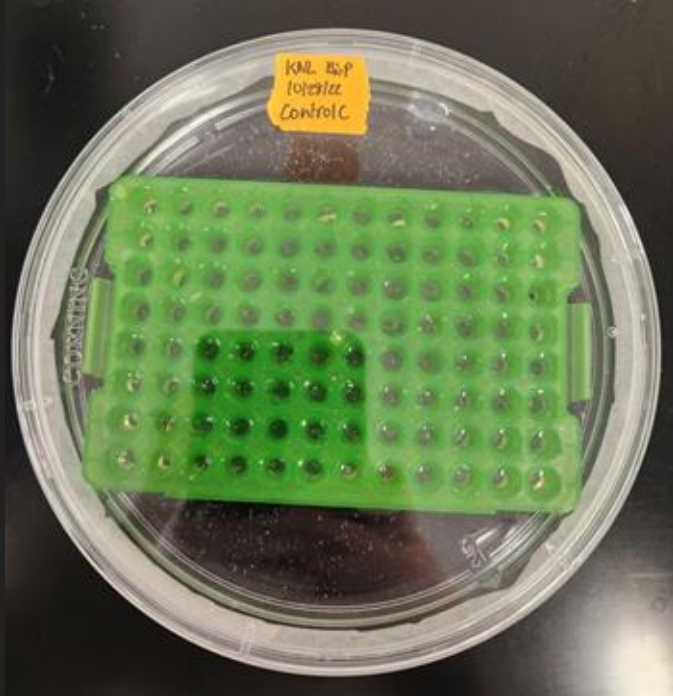
## 28 Oct 2022: Making Control and Experimental Hydroponics

- Control C (on the left) - The bottle of normal John's media was almost used up. There was also small particles remaining in the liquid media in the plate.
- Experimental C (on the right) - Some of the agar was starting to solidify near the end of pouring the agar into the tip racks. This is noticeable at the top right of the image.



## 28 Oct 2022: Making Control and Experimental Hydroponics

- Control C (on the left) - The bottle of normal John's media was almost used up. There was also small particles remaining in the liquid media in the plate.
- Experimental C (on the right) - Some of the agar was starting to solidify near the end of pouring the agar into the tip racks. This is noticeable at the top right of the image.



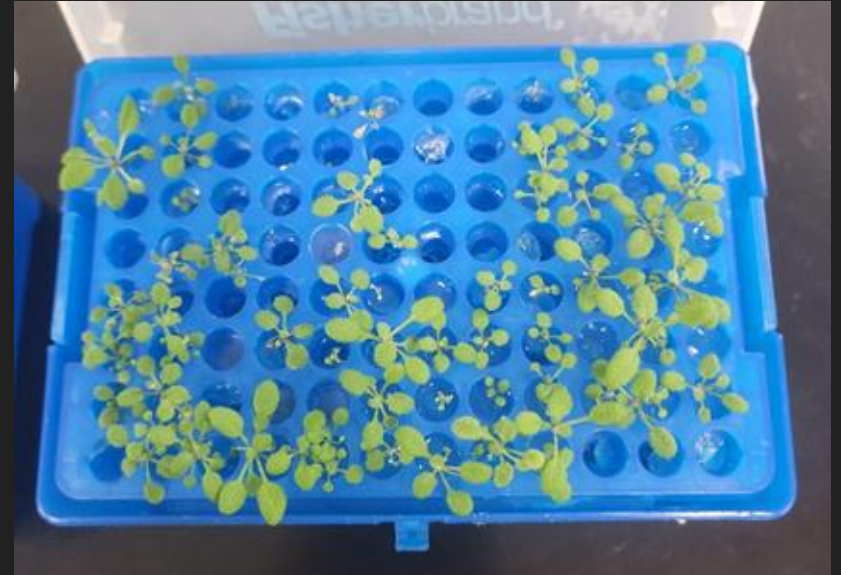
## 31 Oct 2022: Control A (Day 24)

- Majority of the plants died up. Some of the remaining plants grew a lot, but they appear stressed because they grew really tall.



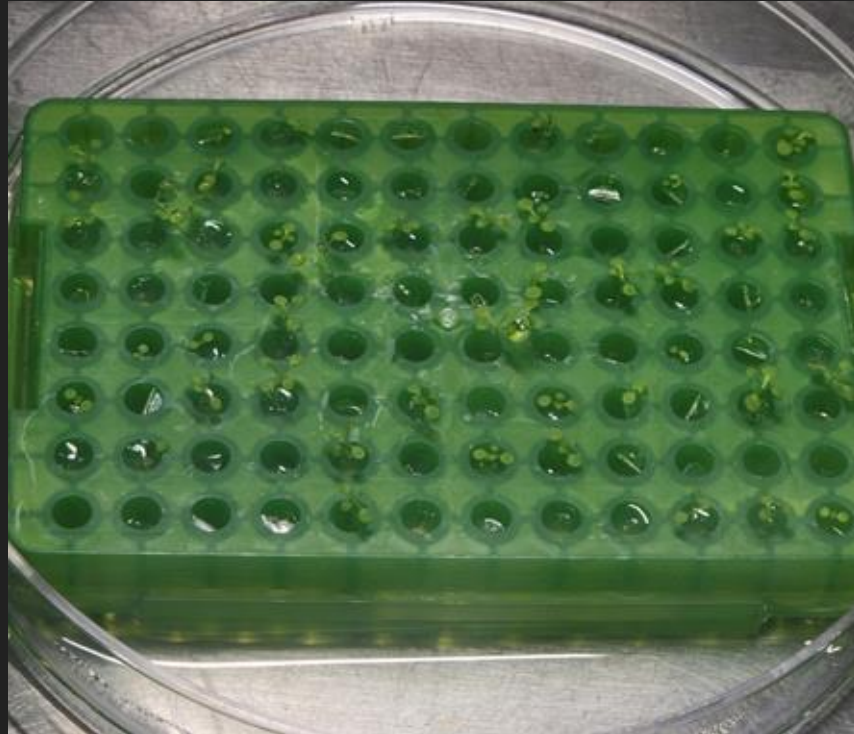
**31 Oct 2022: Control B (Day 18)**

- The plants grew larger after 3 days. The plants at the top middle area dried out.



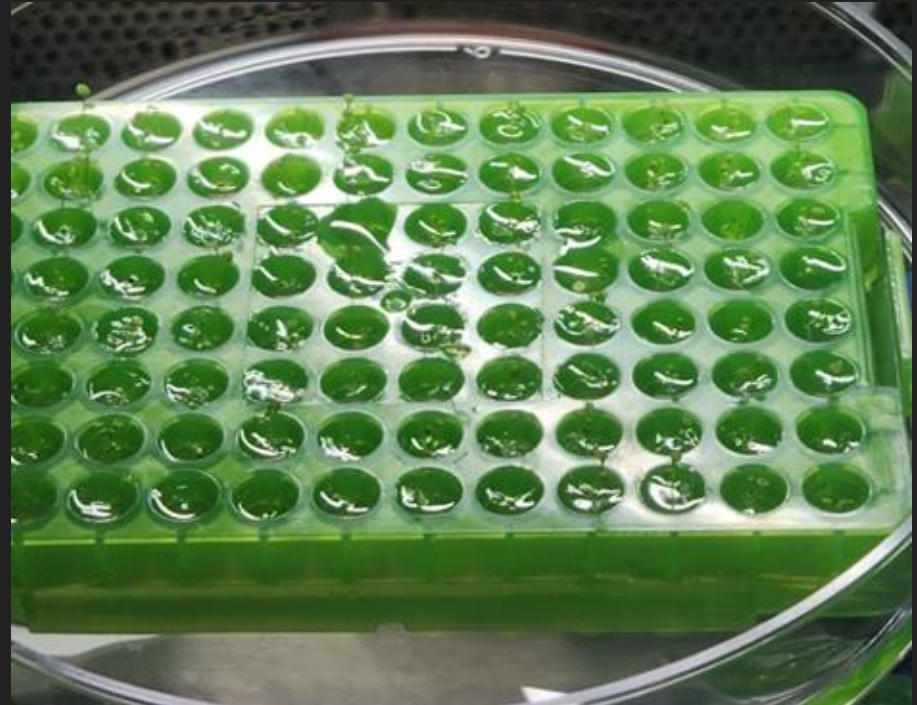
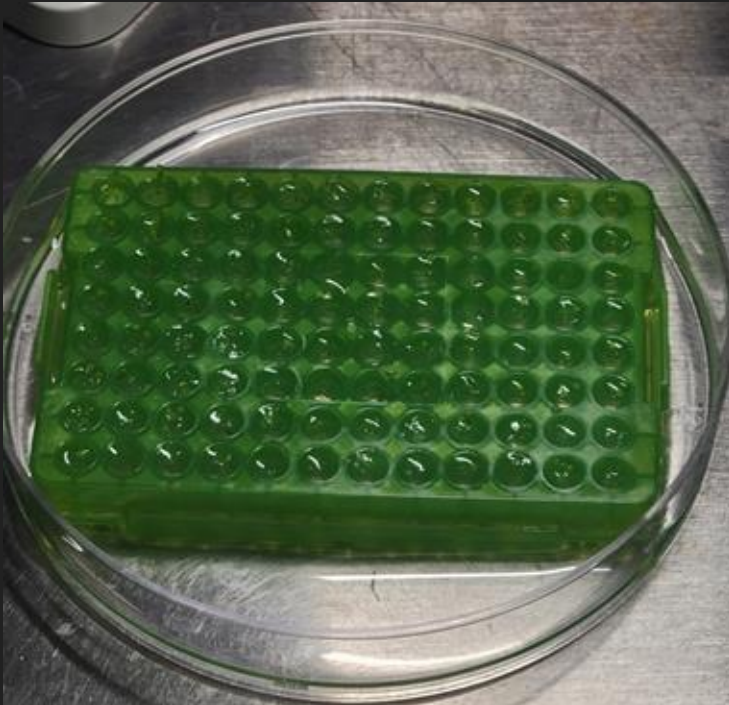
## 31 Oct 2022: Experimental A (Day 10)

- Majority of the seedlings grew more. Some of the seedlings in the middle of the tip rack are being drowned by the John's media that is condensed at the top of the petri plate. However, unlike previous experiences, the plants haven't touched the lid of the petri plate yet.

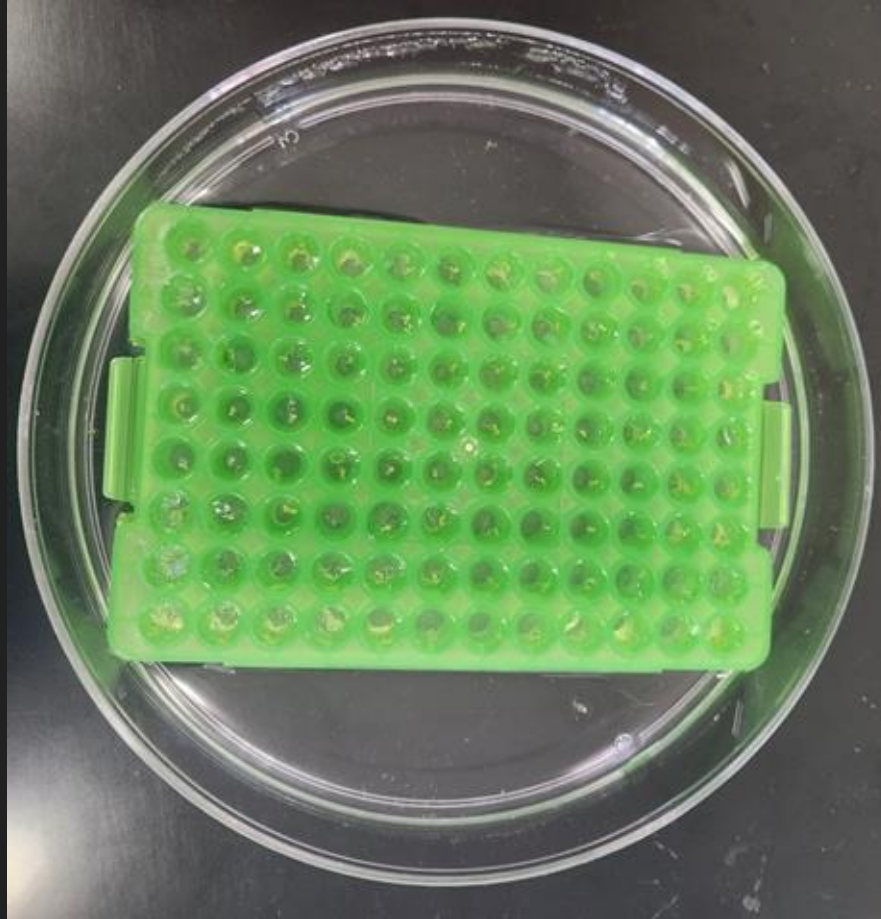


## 31 Oct 2022: Experimental B (Day 6)

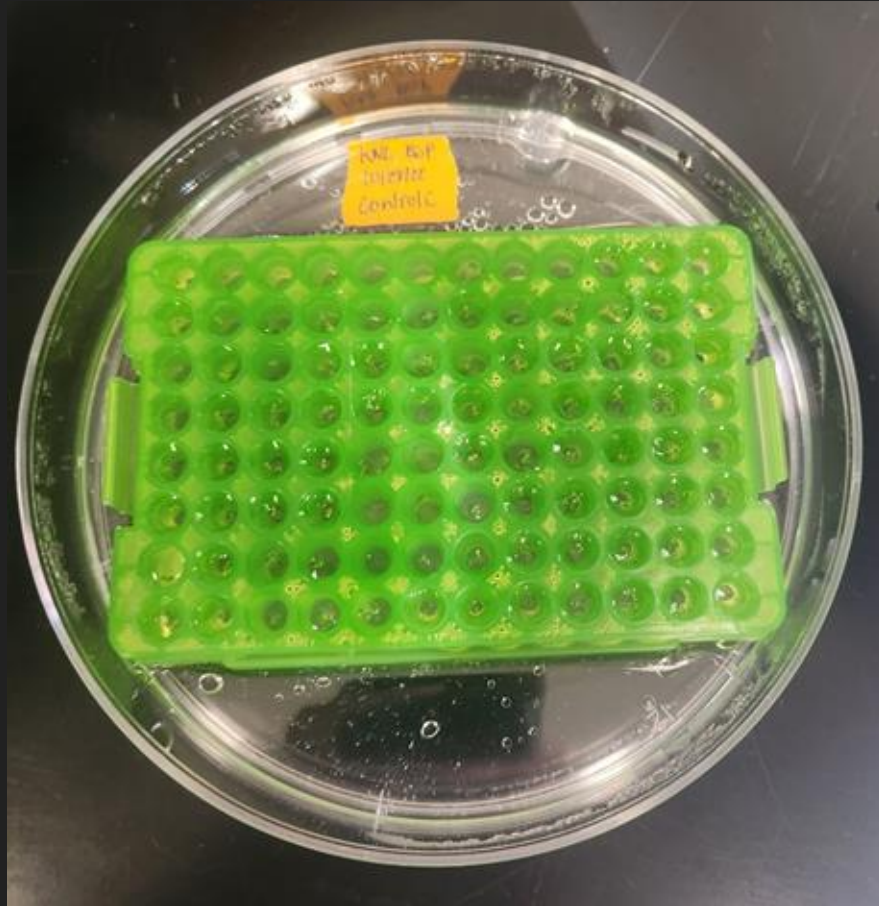
- The seeds are growing/sprouting. Similar to Experimental A, the plants at the middle of the tip rack, are drowned by John's media that was condensed at the lid of the petri plate.



**31 Oct 2022:** Experimental C (Day 3) - The seedlings are starting to germinate.

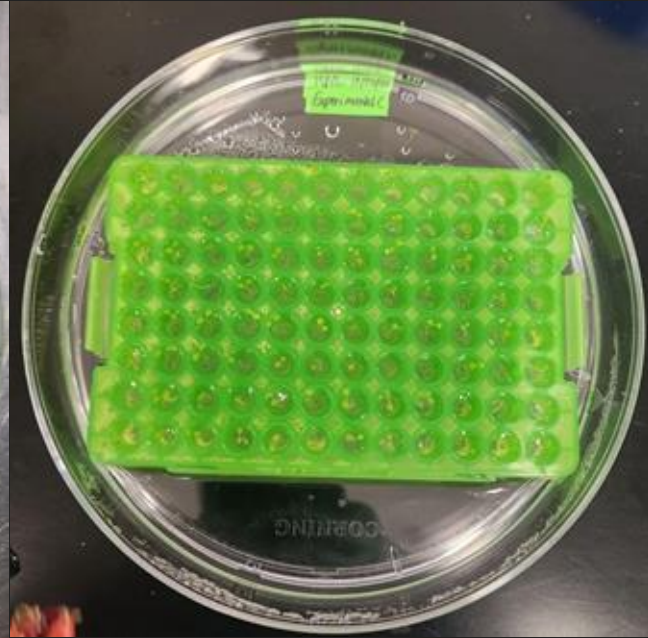
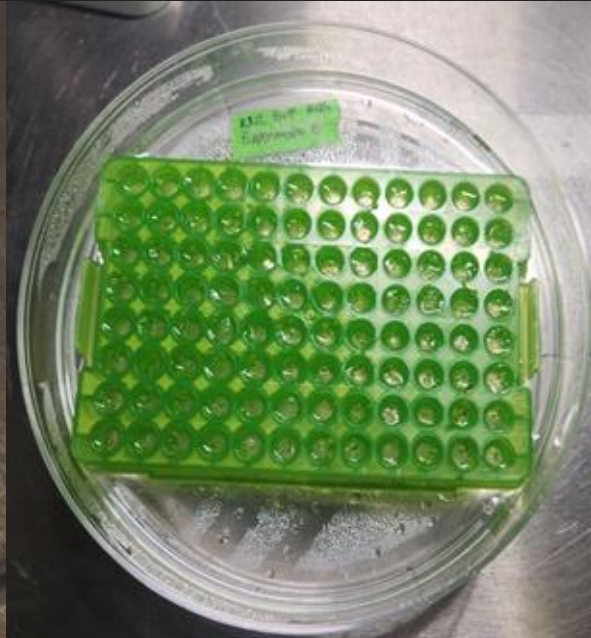
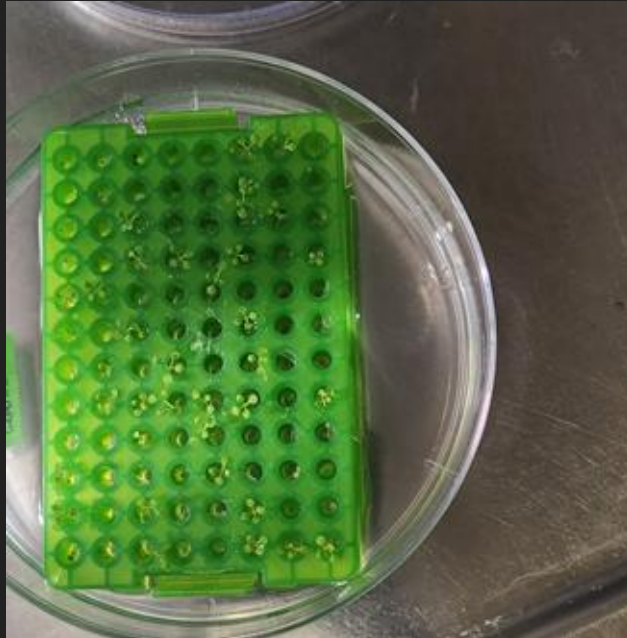


**31 Oct 2022:** Control C (Day 3) - The seedlings are starting to germinate.

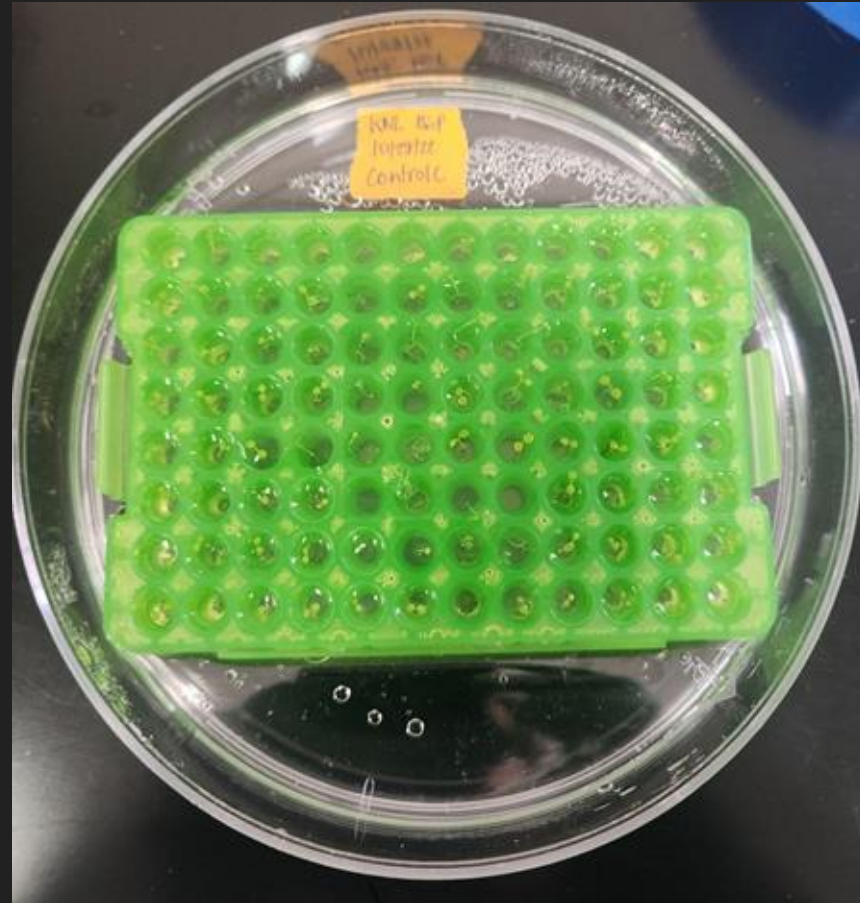


**Nov 1, 2022:**

- Experimental A in control John's media(Day 11 - left): The plants are beginning to touch the lid of the petri plate, and it can be exposed to water stress. As a result, the lid was lifted.
- Experimental B (Day 7 - middle) and Experimental C (Day 4 - right) in John's media with excess copper



**Nov 1, 2022:** Control C (Day 4) - Refilling the petri plate with John's media.



Nov 1, 2022:

- Control A (Day 25 - left): The plants are stressed, which can be identified by the plants that grew very tall.
- Control B (Day 19 - right): The one plant on the far left is very stressed. The plants that are grown at the bottom and the right of the tip rack look healthy and ready for lipid extraction.



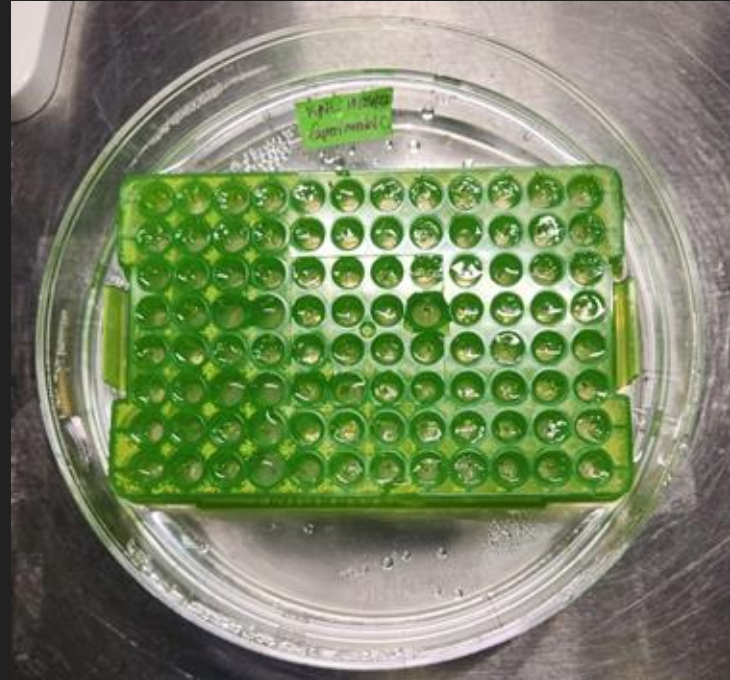
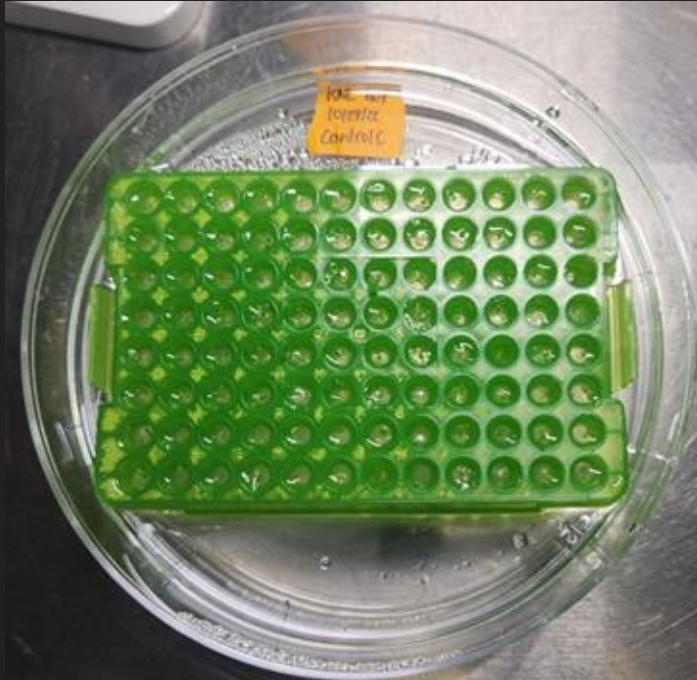
**Nov 3, 2022:**

- Control A (Day 27 - left): Majority of the plants died and the remaining plates are stressed. This is a failed hydroponic; however, it will be utilized as practice for future procedures, such as harvesting plant tissue, freeze drying plant tissue, and lipid extraction.
- Control B (Day 21 - right): The one plant on the far left is very stressed. The plants that are grown at the bottom and the right of the tip rack look healthy and ready for lipid extraction.



**Nov 3, 2022:** The plants in both groups are growing slower than the previous hydroponics.

- Control C (Day 6 - left): Some of the plants that were located in the middle of the tip rack were drowned, so the lid was lifted for a couple of hours.
- Experimental C (Day 6 - right): The plants are growing a lot. Some of the plants are approaching the lid of the petri plate.



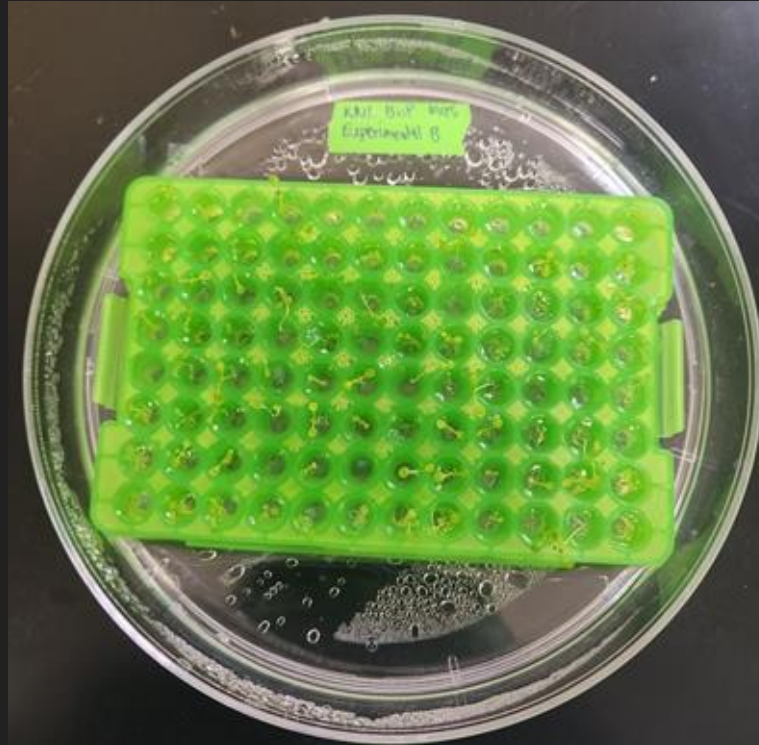
**Nov 3, 2022:**

- Experimental A (Day 13): The tip rack was transferred from the petri plate to the tip boxes and the John's media with excess copper was utilized. The plants look healthy, but have a paler green color compared to the negative control plants.



**Nov 3, 2022:**

- Experimental B (Day 9): The plants are drowned from the condensed John's media, in, which some of them are tilting to the side. A few of the plants are starting to touch the lid of the petri plate, so the tape is taken off and the lid is lifted.

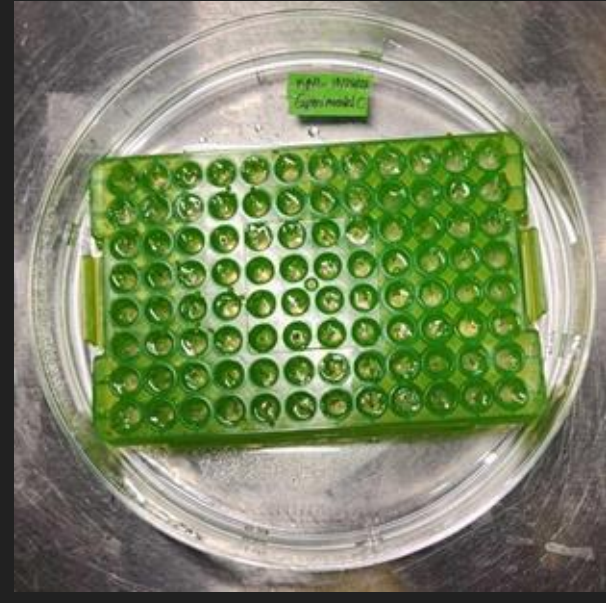
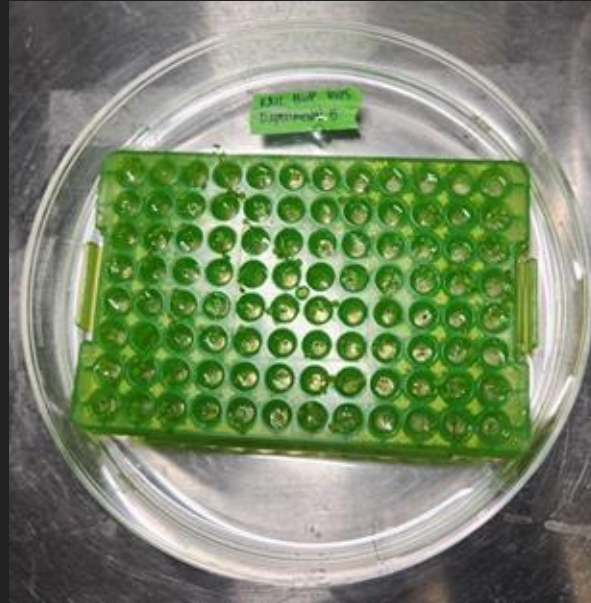
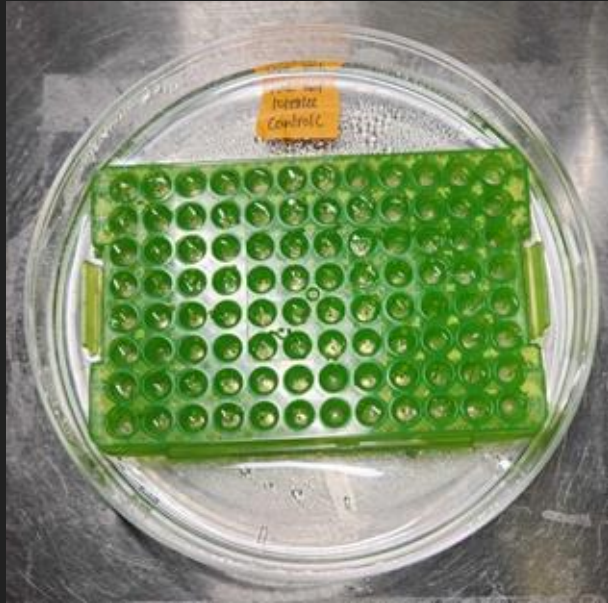


**Nov 3, 2022:**

- Samples submerged in liquid N<sub>2</sub> to freeze dry the Arabidopsis plant (left).
- 17 Freeze dried samples are located in the bottom section of the -80°C freezer (located in the corner of the room, next to the printer.)

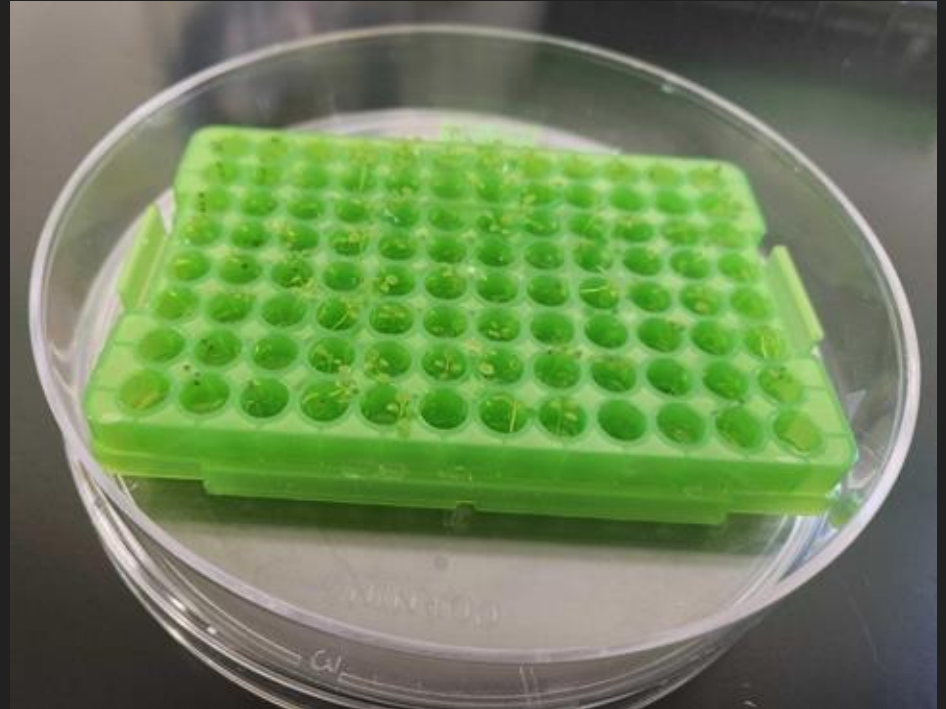
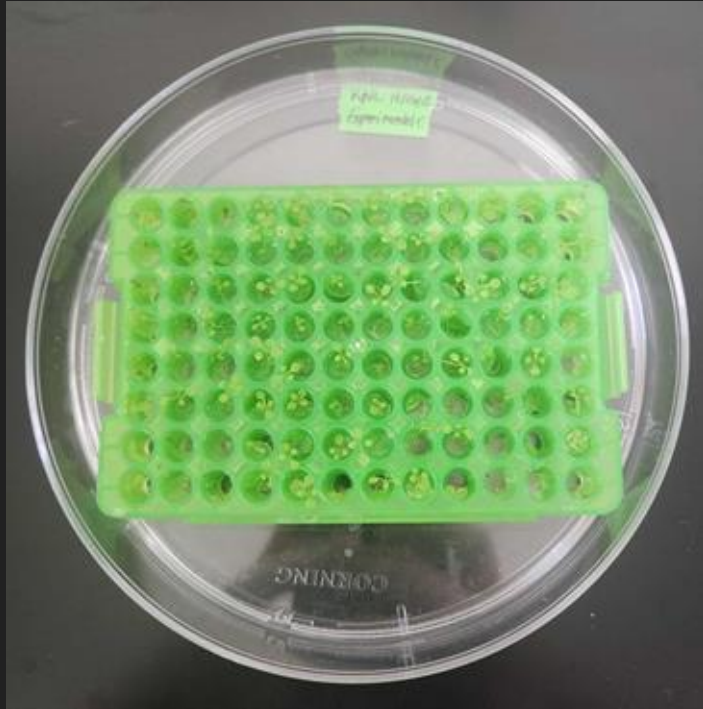


**Nov 4, 2022:** Refill/Replacing media in hydroponics: Control C (left), Experimental B (middle), Experimental C (right). All of the plants in three of these hydroponics were drowned with the media that was condensed in the petri plates. Few of the plants in these hydroponics were also approaching or on the lid of the petri plate. As a result, the lids were lifted.



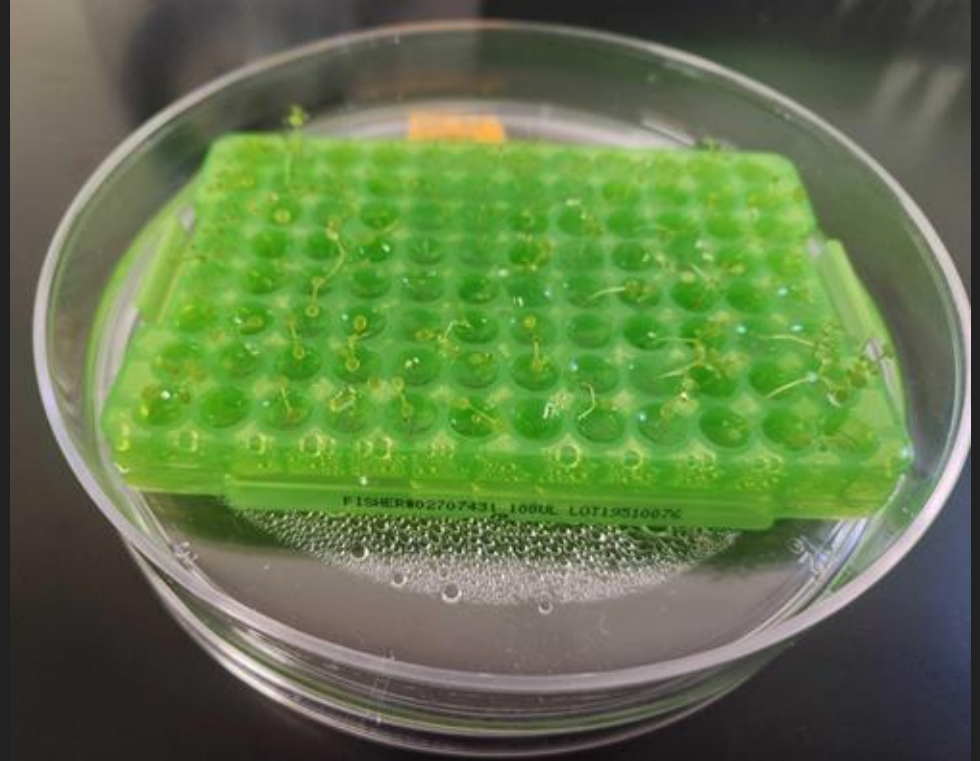
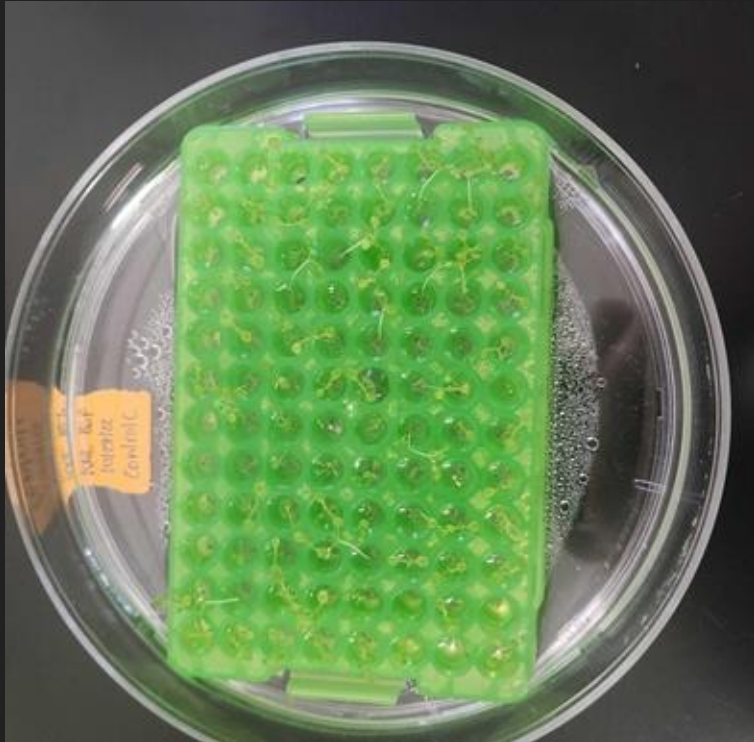
**Nov 7, 2022:**

- Experimental C (Day 10) - Because the lid was lifted over the weekend, all of the media as well as some of the plants were dried out. The plate was refilled with normal John's media, closed, and sealed.



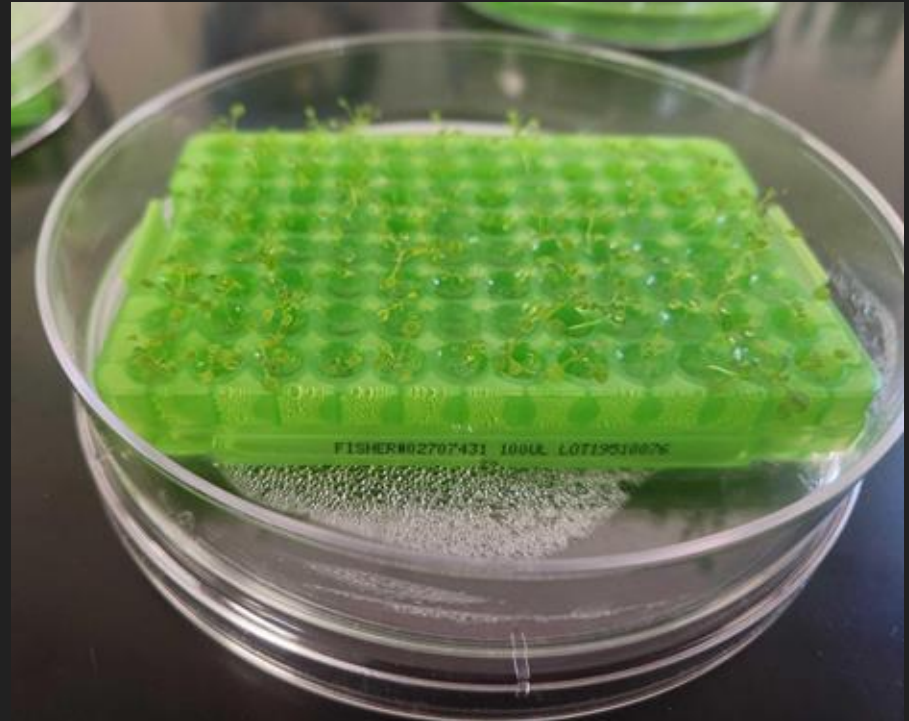
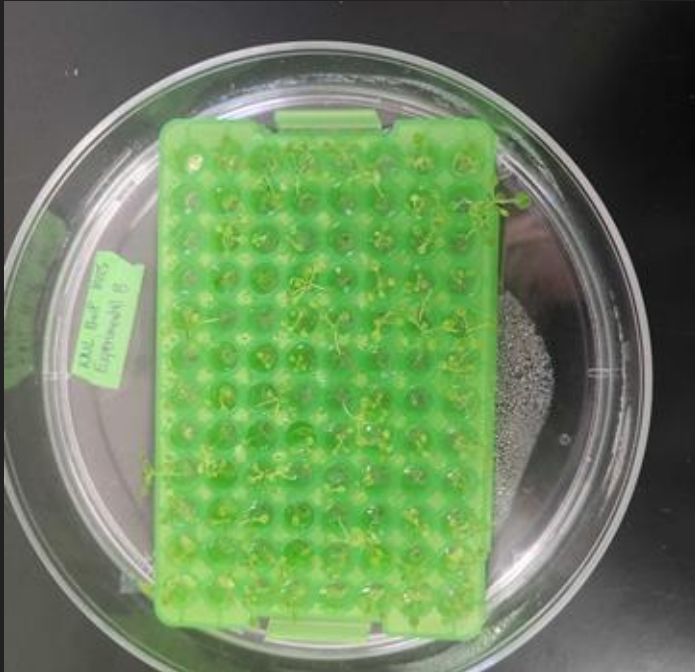
**Nov 7, 2022:**

- Control C (Day 10): Someone closed the lid, so the plants were tilted and drowned by the condensed John's media. Since there were no more autoclaved tip boxes, the lid was lifted for a day to allow the media on the plants to dry out.



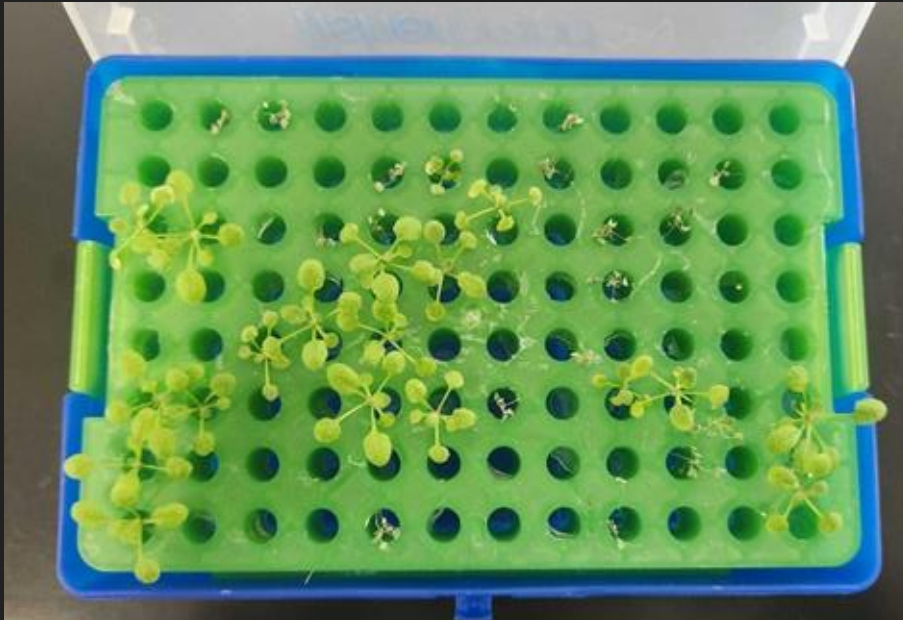
**Nov 7, 2022:**

- Experimental B (Day 13): Someone closed the lid, so the plants were tilted and drowned by the condensed John's media. Since there were no more autoclaved tip boxes, the lid was lifted for a day to allow the media on the plants to dry out.



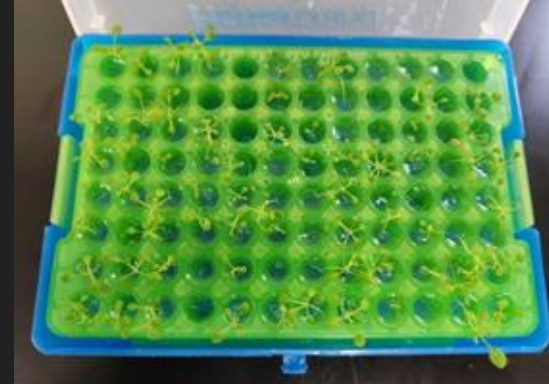
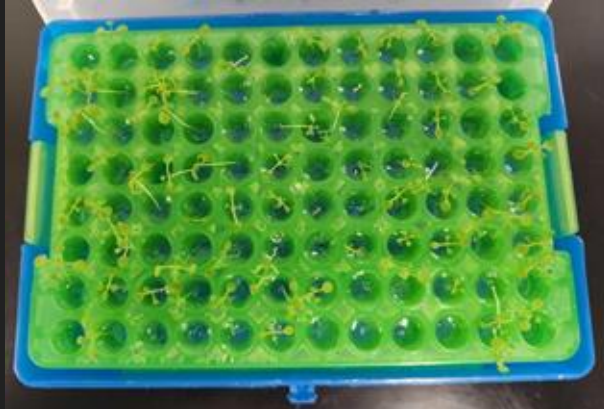
**Nov 7, 2022:**

- Experimental A (Day 17): The media evaporated, and was no longer touching the roots of the small plants. As a result, these plants dried out and died.



Nov 8, 2022:

- Control C (left) and Experimental B (right): The tip racks were moved from the petri plates to the autoclaved tip boxes.



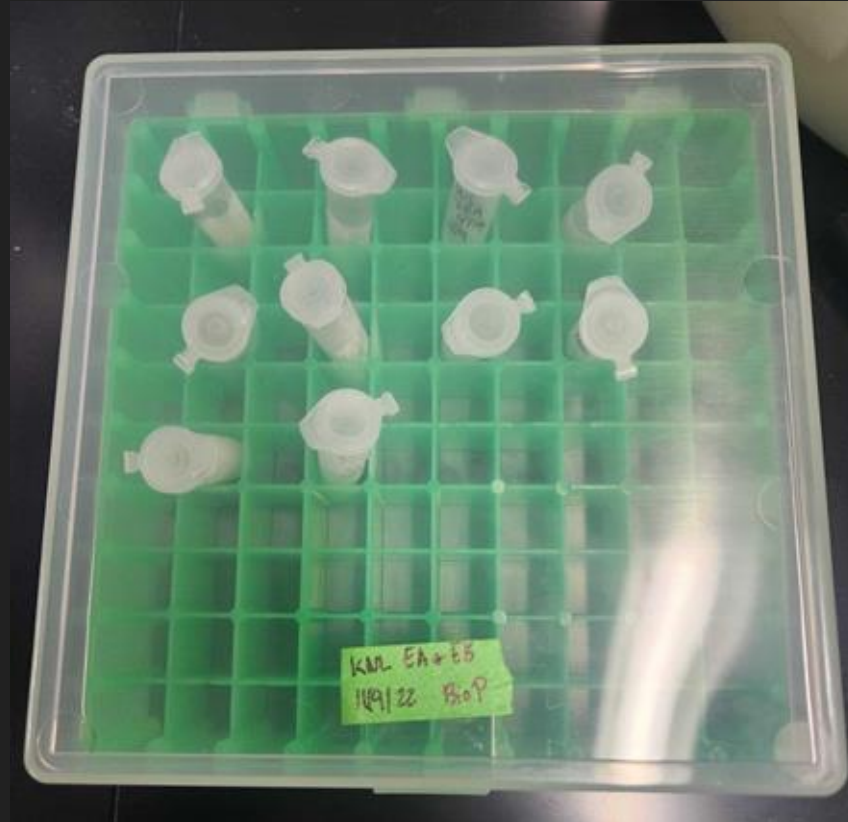
Nov 8, 2022:

- MTBE/MeOH (10/3 v/v) (left vial)
- MTBE/MeOH/H<sub>2</sub>O (10/3/2.5 v/v/v) (right vial)



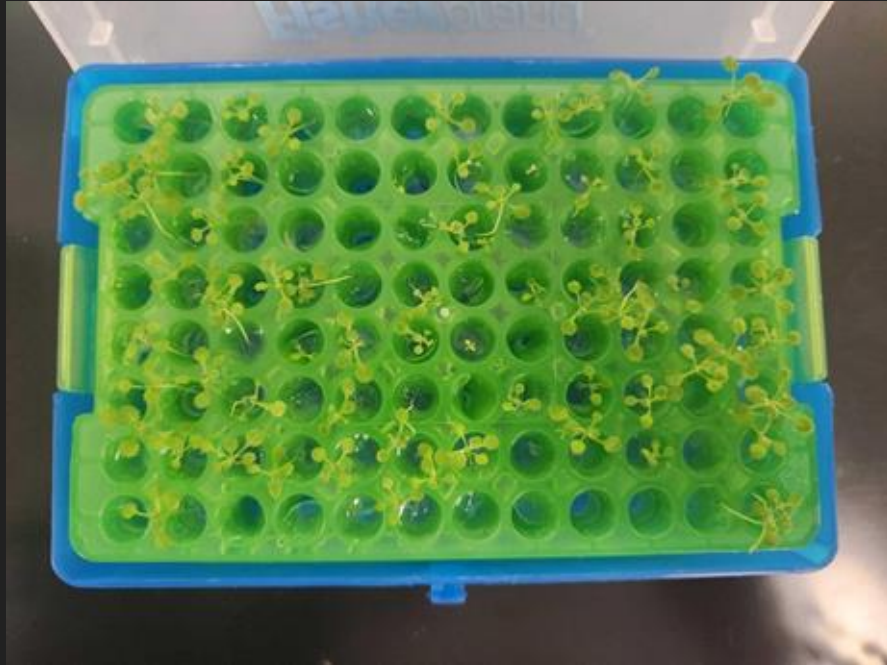
**Nov 9, 2022:**

- Freeze dried Experimental A and Experimental B plants



**Nov 10, 2022:**

- Control C (left) and Experimental C (changed to normal) (right)
  - Control taller than Experimental C



## Nov 10, 2022: Liquid-Liquid Extraction (From left to right)

1. T2CB (left vial) and T4CB (right vial) after 20 minutes of shaking lipids and MTBE/MeOH on ice.
2. T2CB (left vial) and T4CB (right vial) after 20 minutes of shaking lipids and MTBE/MeOH/H<sub>2</sub>O on ice.
3. Collection of lipids that were extracted from T2CB (left vial) and T4CB (right vial).



**Nov 14, 2022:**

- Experimental C (left) and Control C (right)
  - Control has more plant biomass than Experimental C.

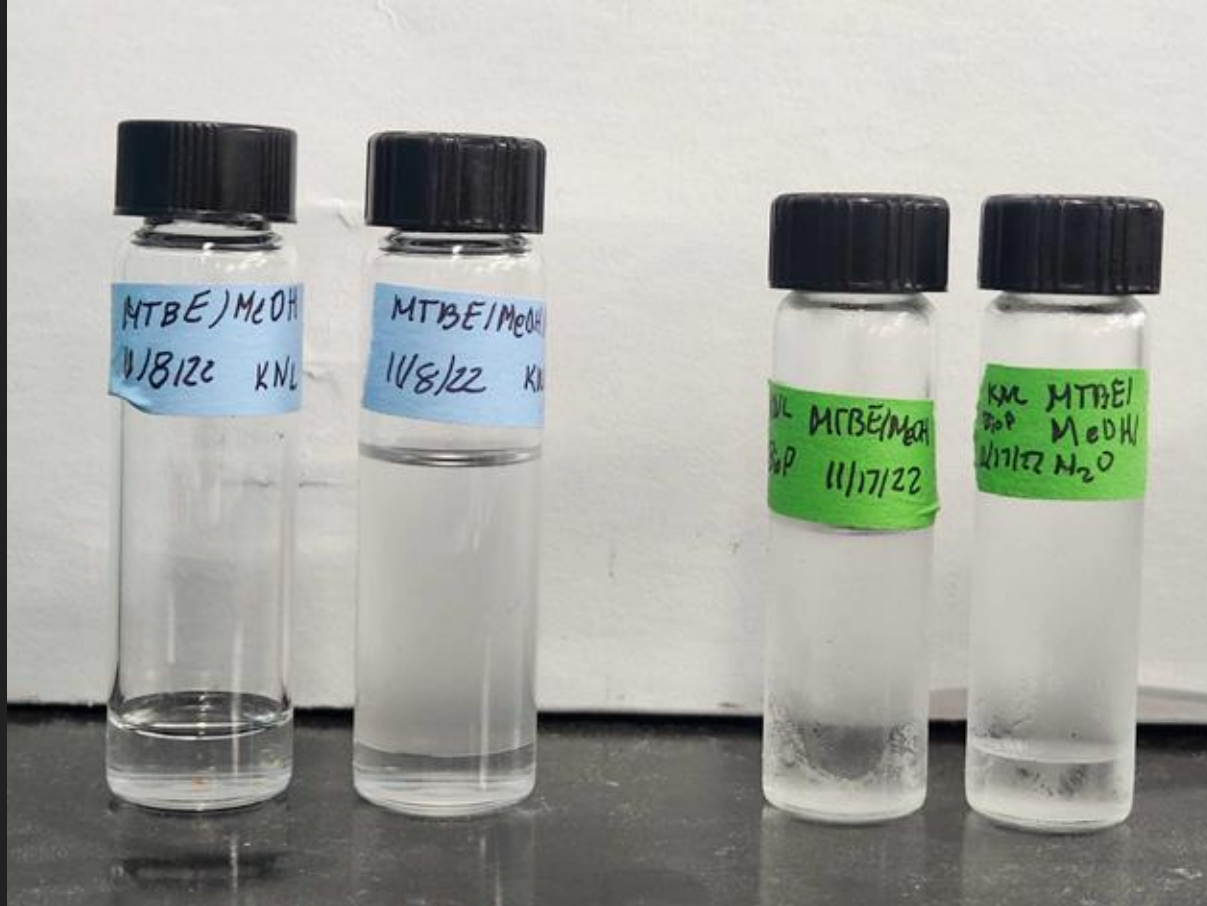


## Nov 16, 2022: Liquid-Liquid Extraction (From left to right)

1. T1EA (left vial), T2EA (middle vial), and T1CB (right vial) before solution is shaken on ice.
2. T1EA (left vial), T2EA (middle vial), and T1CB (right vial) after 20 minutes of shaking lipids and MTBE/MeOH on ice.
3. T1EA (left vial), T2EA (middle vial), and T1CB (right vial) after 20 minutes of shaking lipids and MTBE/MeOH/H<sub>2</sub>O on ice.
4. Collection of lipids that were extracted from T1EA (left vial), T2EA (middle vial), and T1CB (right vial).



**Nov 17, 2022:** Preparing MTBE/MeOH (10/3 v/v) and MTBE/MeOH/H<sub>2</sub>O (10/3/2.5)



## Nov 17, 2022: Experimental and Control Group C on Day 20

- Left: The experimental group has yellowish, dried spots on the leaves. There is also less plant biomass.
- Middle: The control group continues to have green leaves. There is more plant biomass.
- Right: Freeze dried samples. (T28-T42)

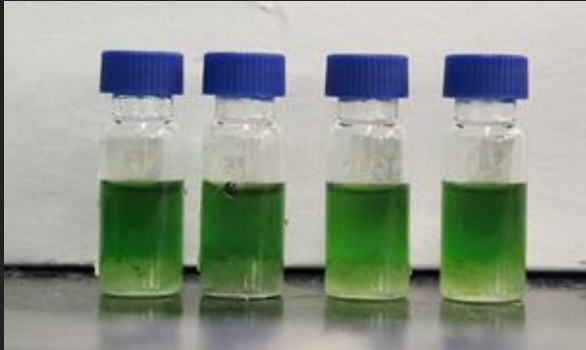


## Nov 17, 2022: Liquid-Liquid Extraction

Samples from left to right: T3CB, T5CB, T3EA, T4EB

1. Left: Vials after 20 minutes of shaking lipids and MTBE/MeOH on ice.
2. Middle: Vials after 20 minutes of shaking lipids and MTBE/MeOH/H<sub>2</sub>O on ice.
3. Right: Collection of lipids that were extracted

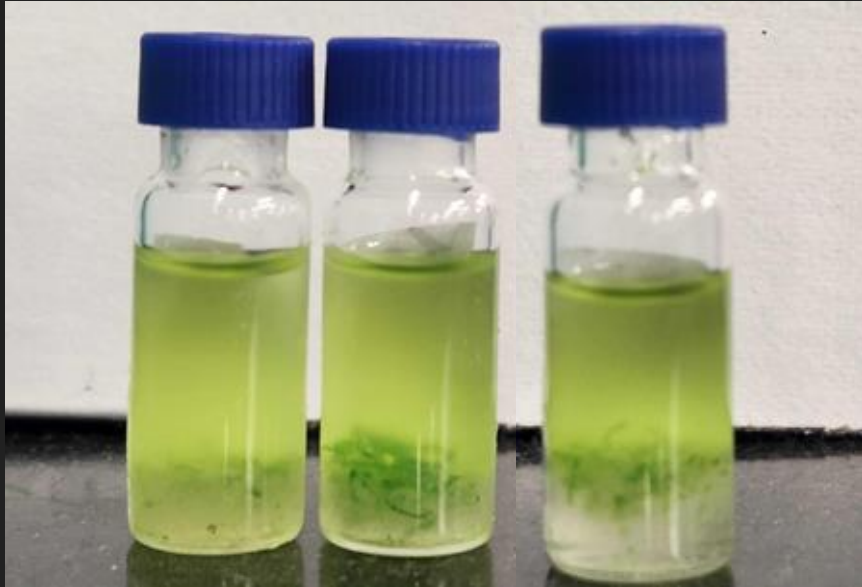
More lipids are extracted when the remaining plant is included.



## Nov 17, 2022: Liquid-Liquid Extraction

Samples from left to right: T6CB, T5EB, T6EB

1. Left: Vials after 40 minutes of shaking lipids and solutions on ice.
2. Right: Collection of lipids that were extracted



**Nov 18, 2022:** Liquid-Liquid Extraction (Samples from left to right: T7CB, T28-T42)

1. Top: Vials after 20 minutes of shaking lipids and MTBE/MeOH on ice.
2. Middle: Vials after 20 minutes of shaking lipids and MTBE/MeOH/H<sub>2</sub>O on ice.
3. Bottom: Collection of lipids that were extracted

